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(54) **BUILDING AUTOMATION SYSTEM WITH INTEGRATED BUILDING INFORMATION MODEL**

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CPC **G05B 13/04** (2013.01); **G05B 15/02** (2013.01); **G05B 2219/25011** (2013.01); **G05B 2219/2642** (2013.01)

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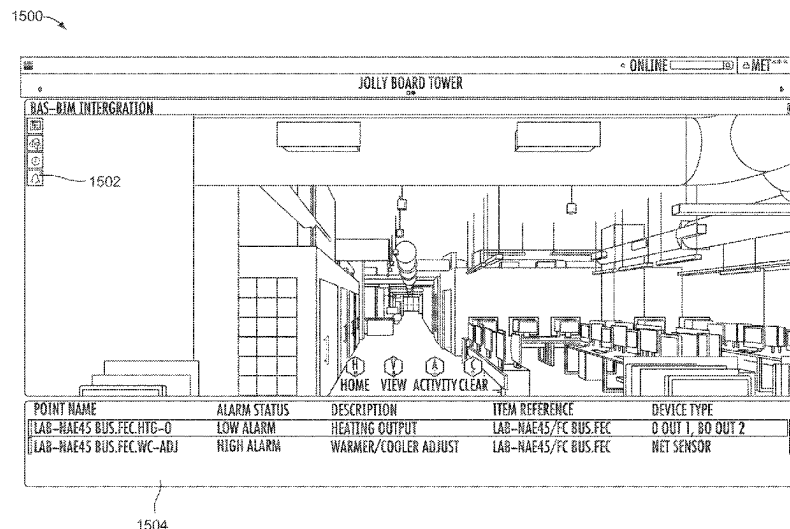
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(57) **ABSTRACT**

A building automation system (BAS) includes building equipment located within a building and a BAS network configured to facilitate communications between the building equipment. The building equipment operate to affect a variable state or condition within the building. The BAS includes a BAS-BIM integrator configured to receive BAS points from the BAS network and to integrate the BAS points with a building information model (BIM). The BIM includes a plurality of BIM objects representing the building equipment. The BAS includes an integrated BAS-BIM viewer configured to use the BIM with the integrated BAS points to generate a user interface. The user interface includes a graphical representation of the BIM objects and the BAS points integrated therewith.

20 Claims, 18 Drawing Sheets



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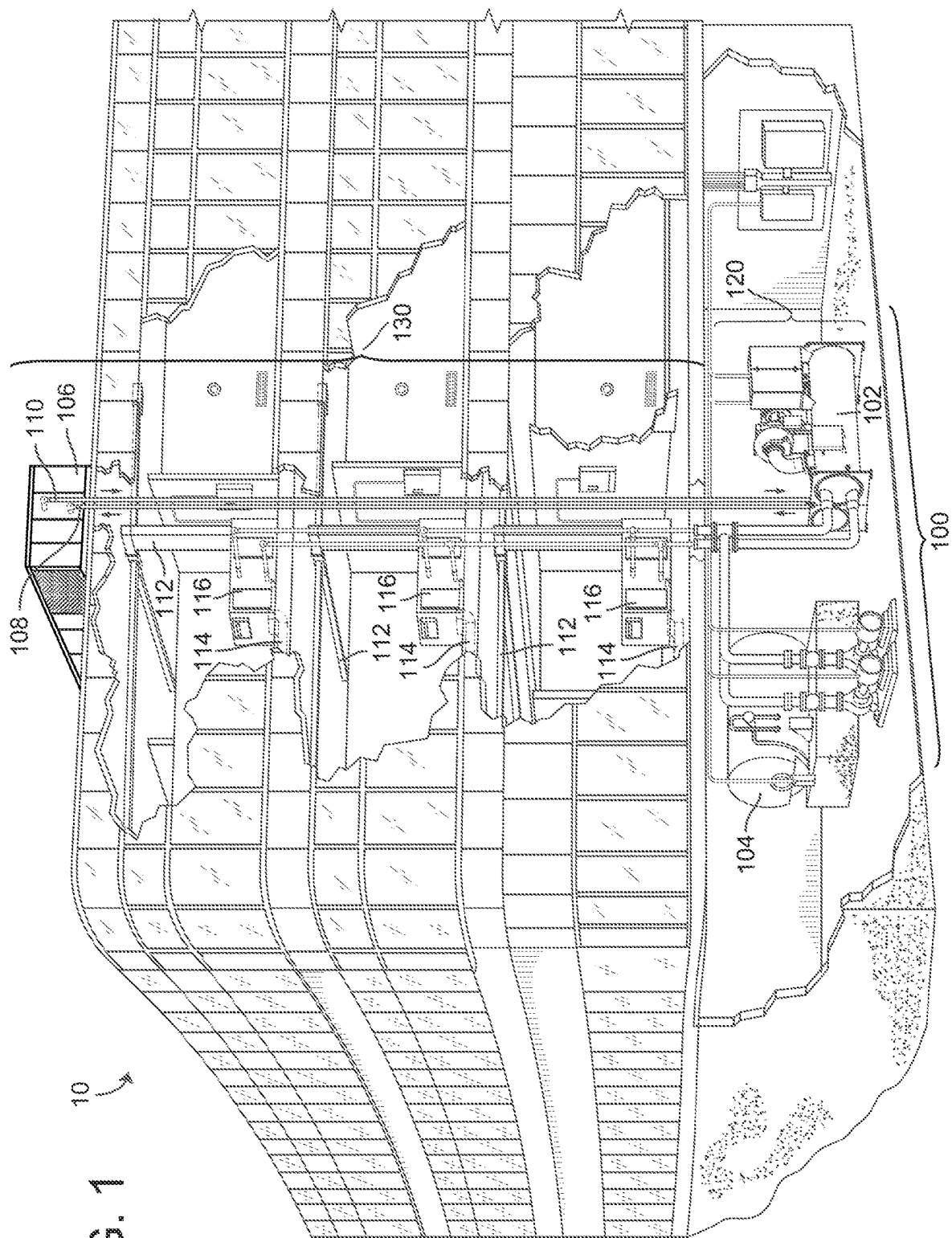
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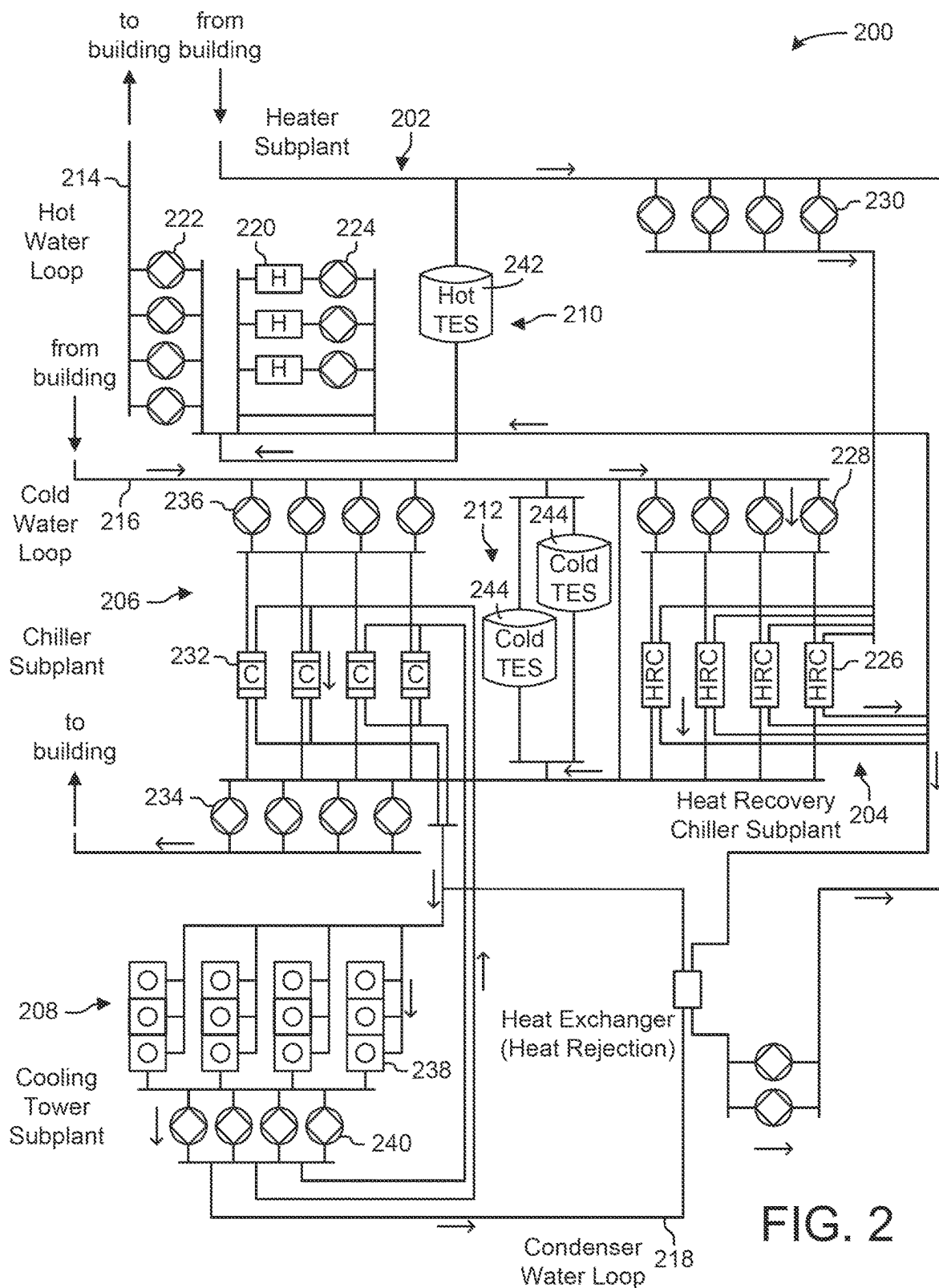


FIG. 2

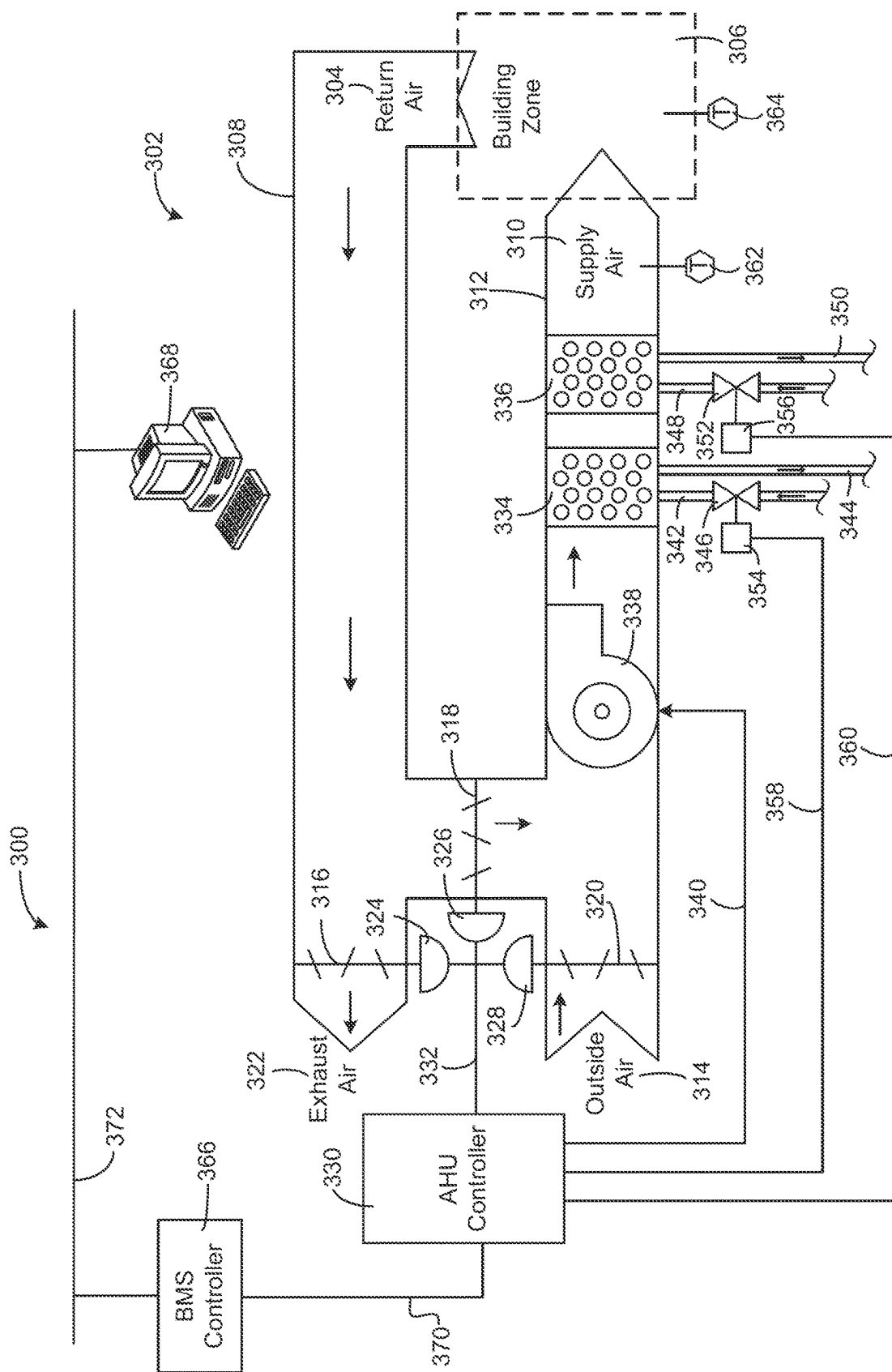
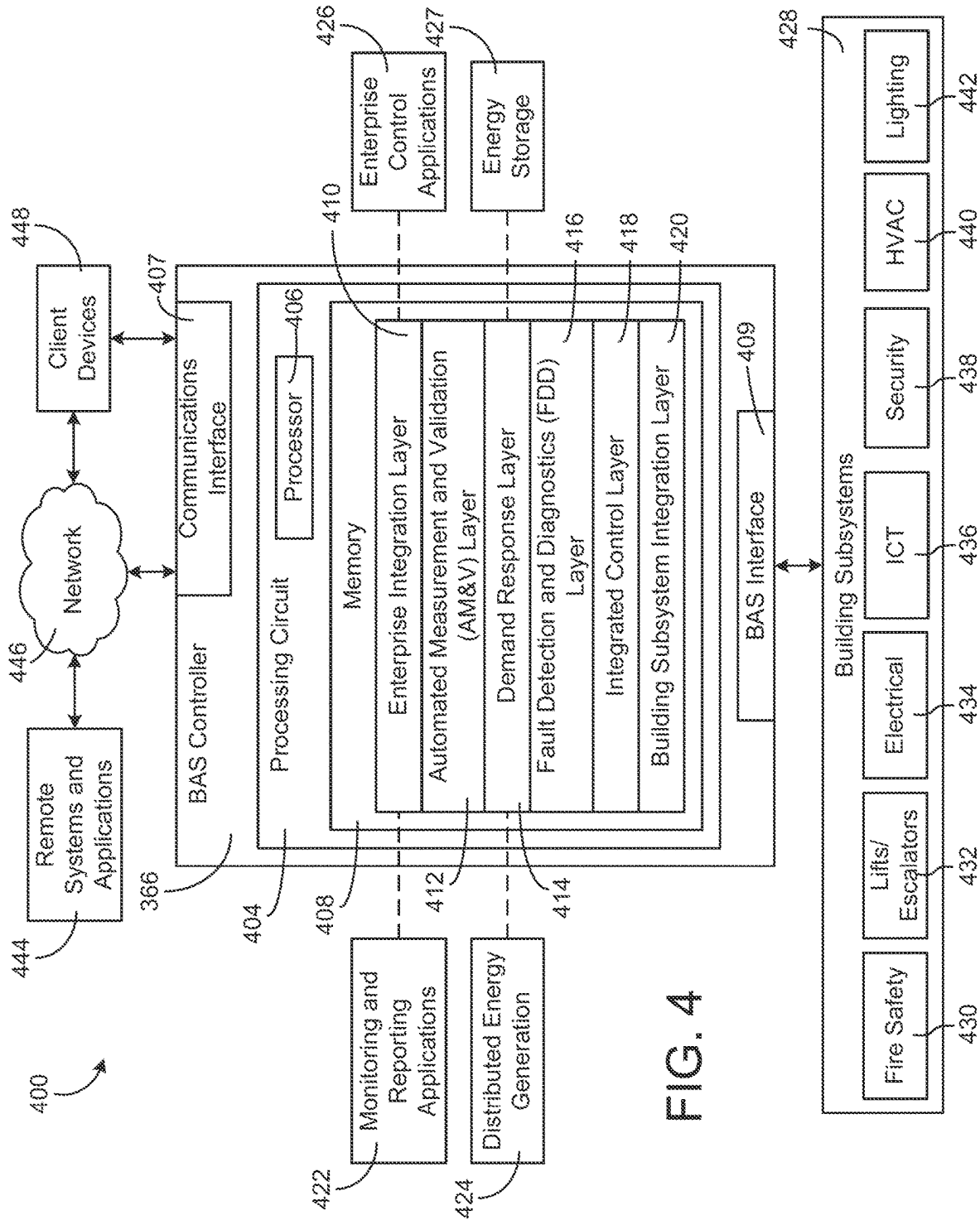
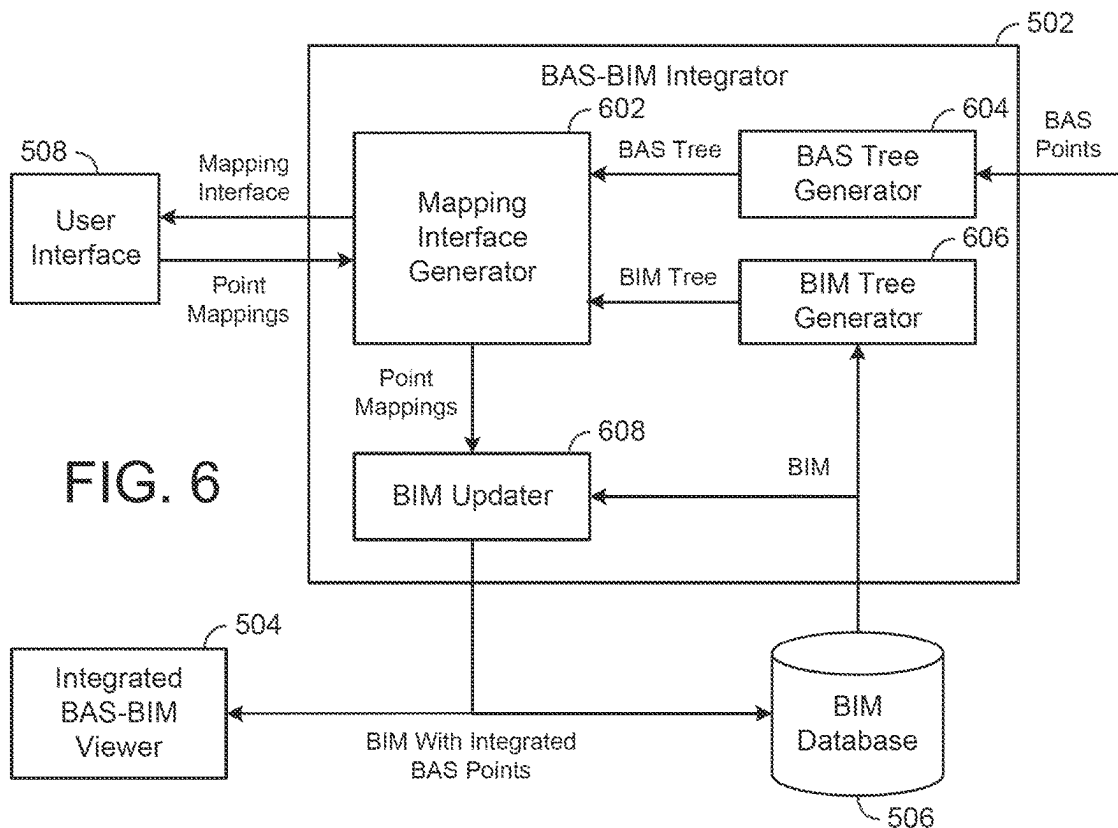
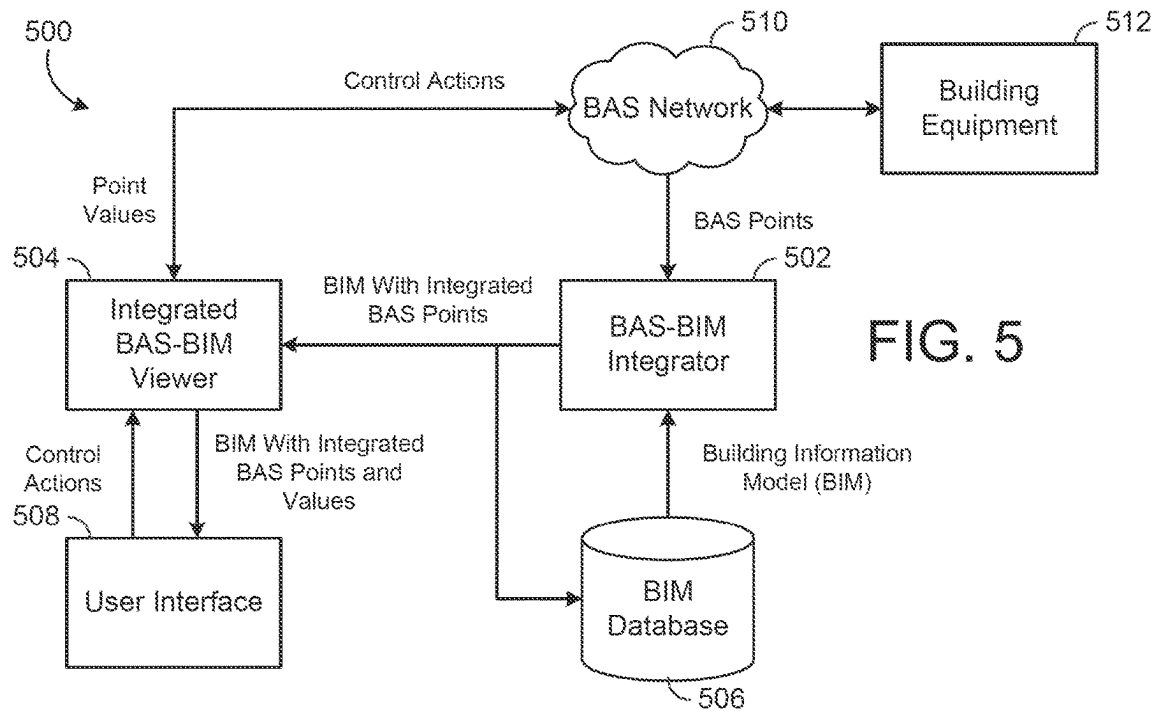


FIG. 3





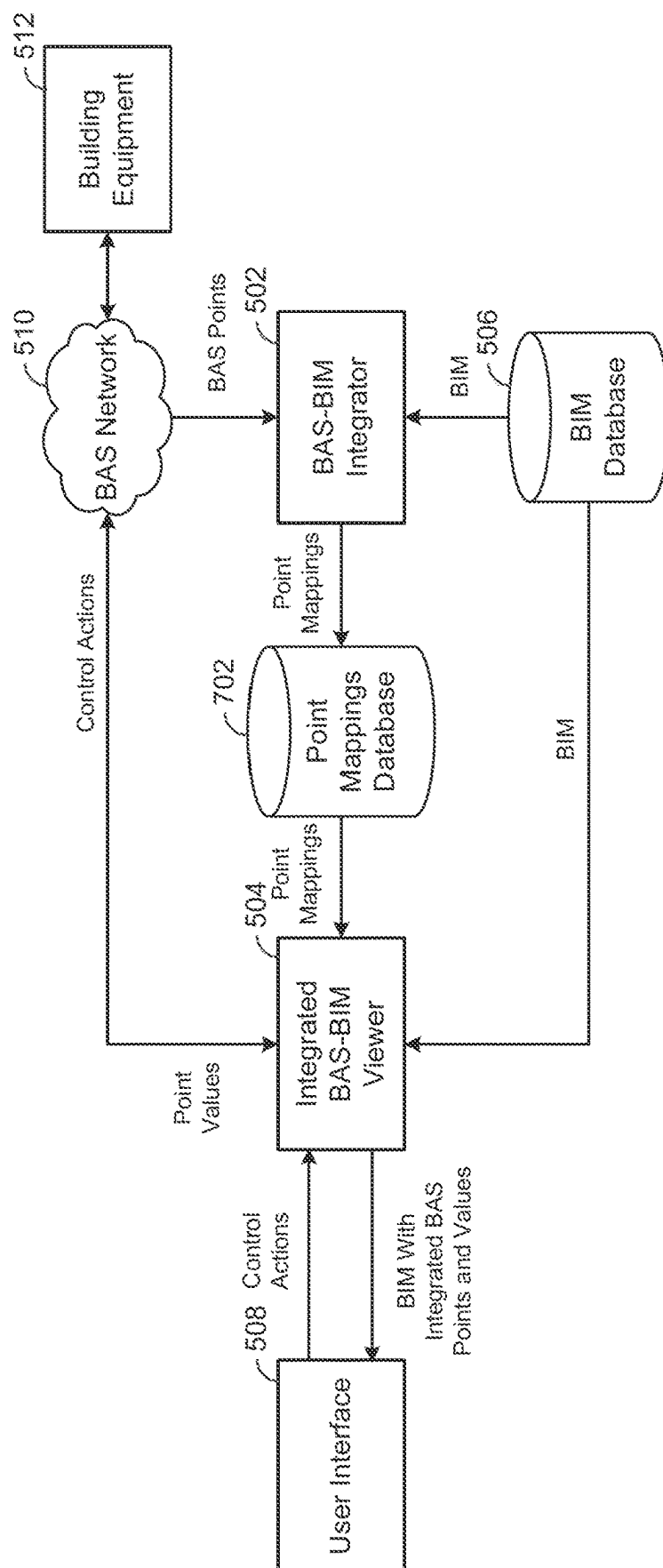
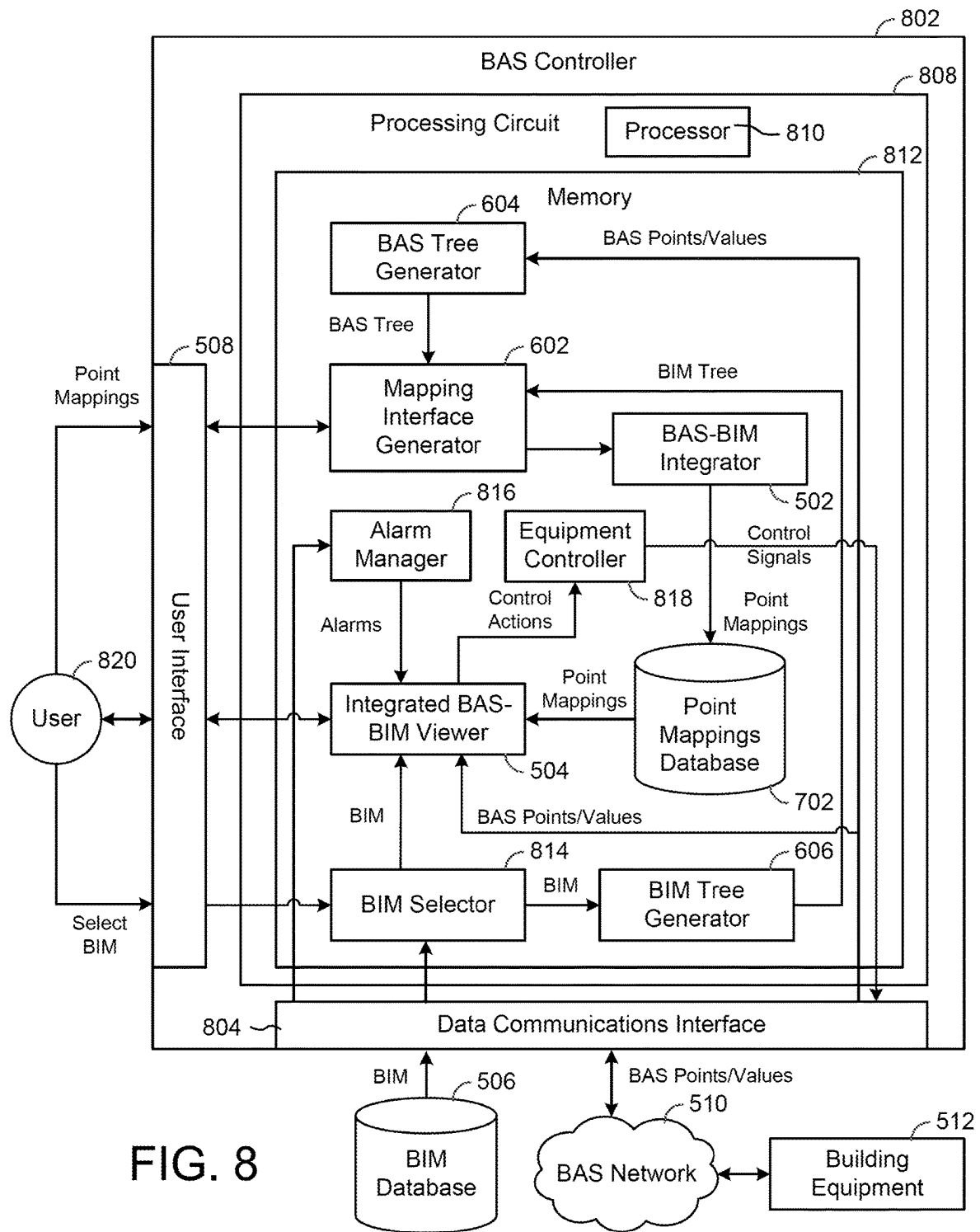


FIG. 7



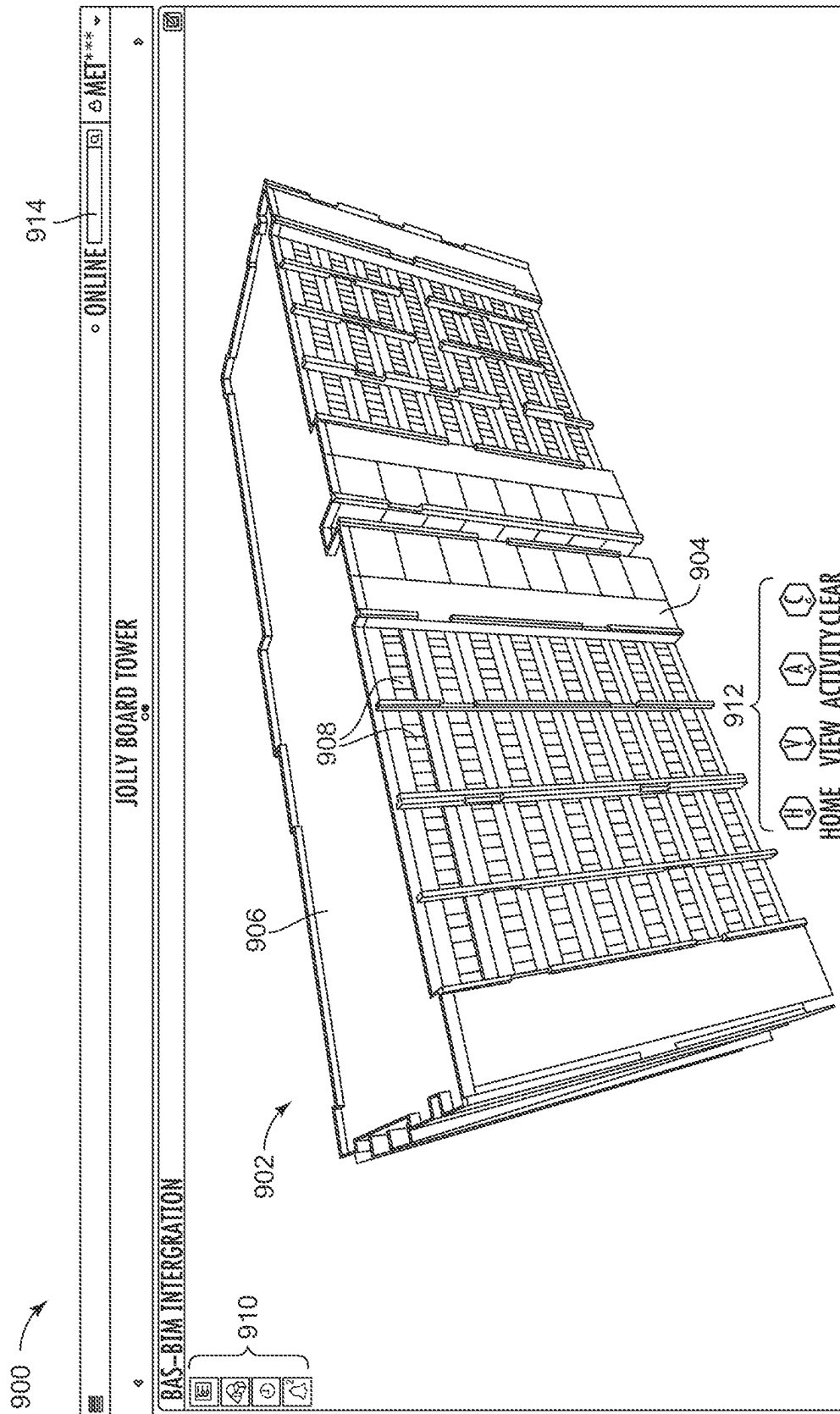
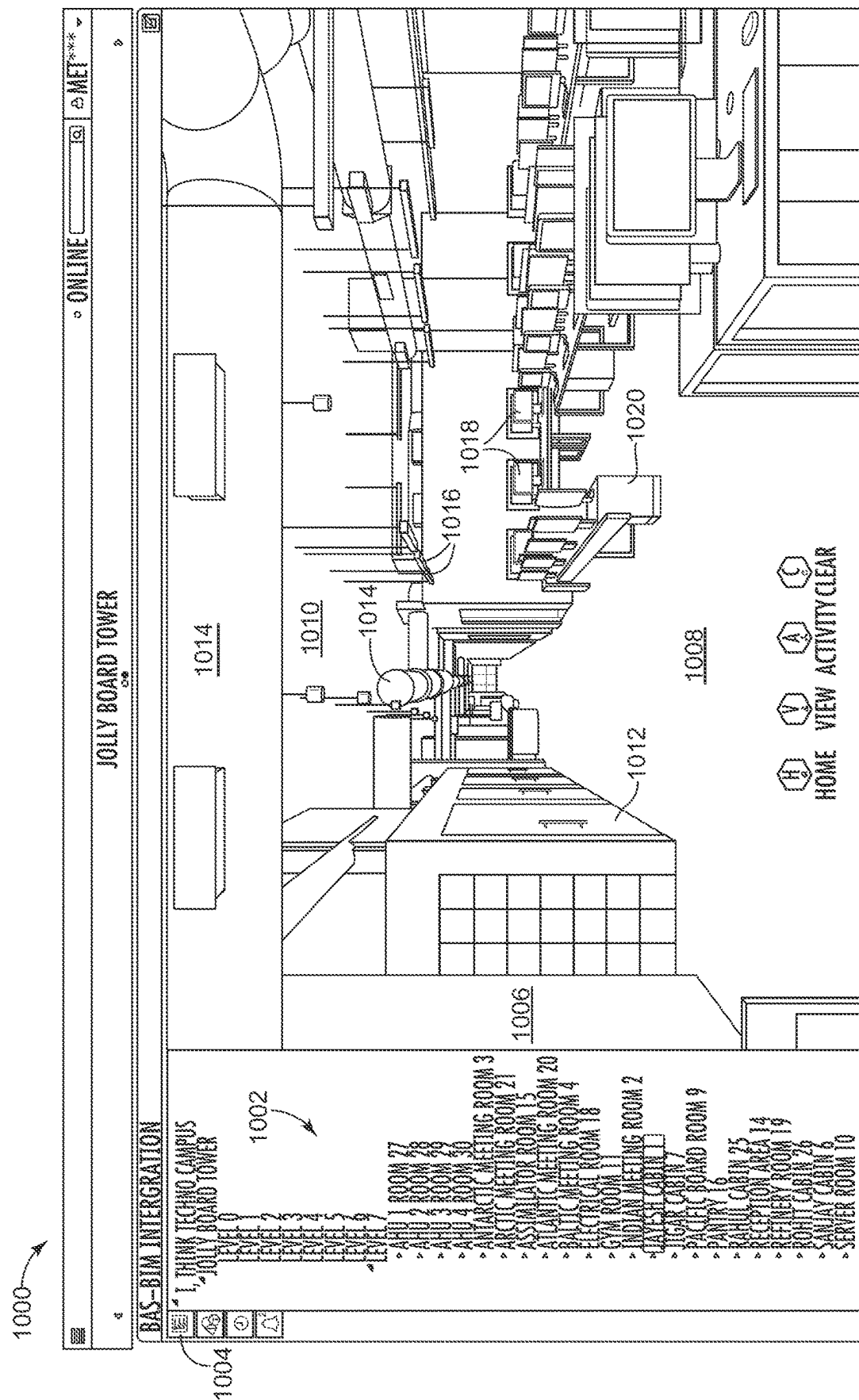


FIG. 9



OF
G^{*}
L

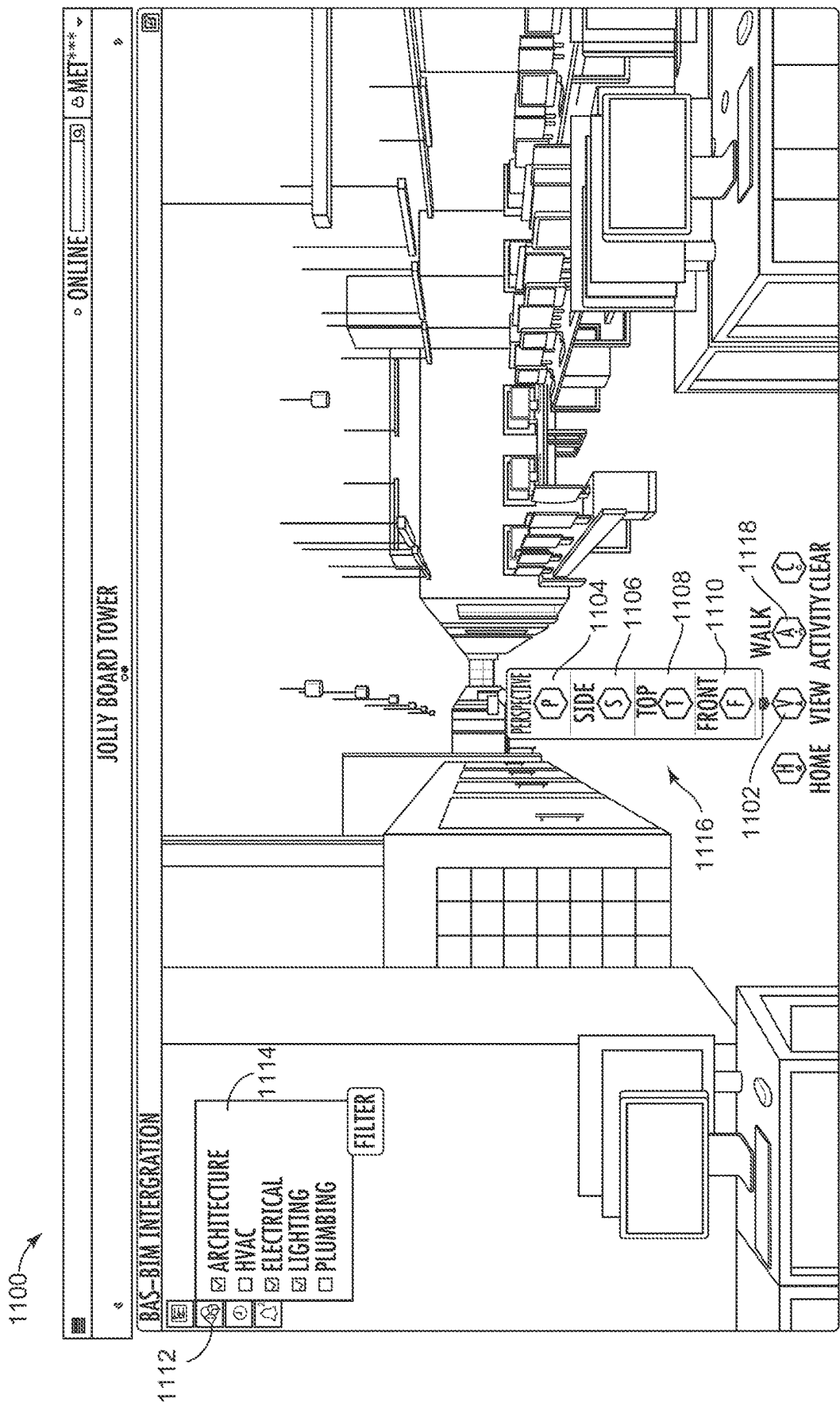


FIG. 11

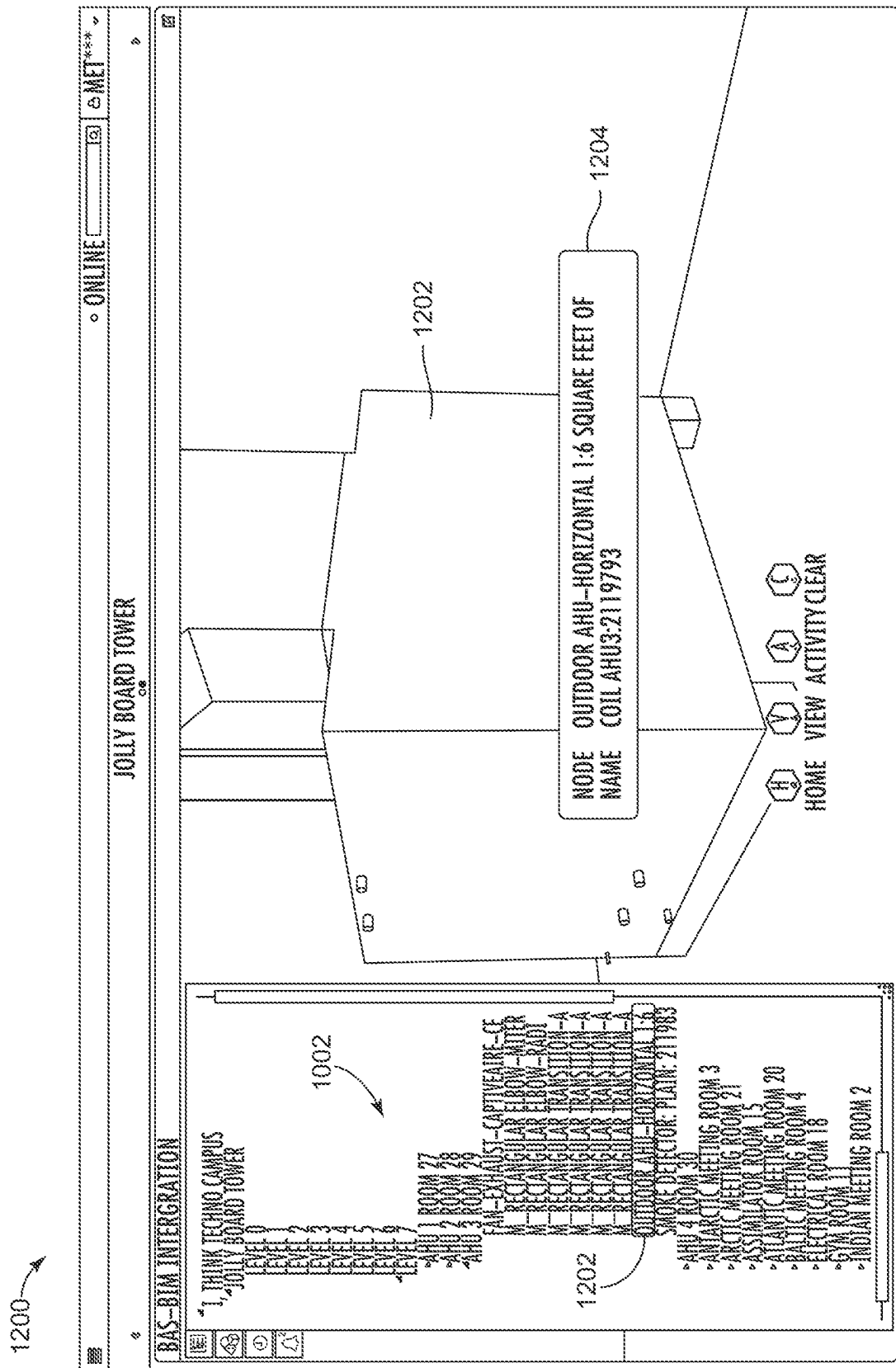
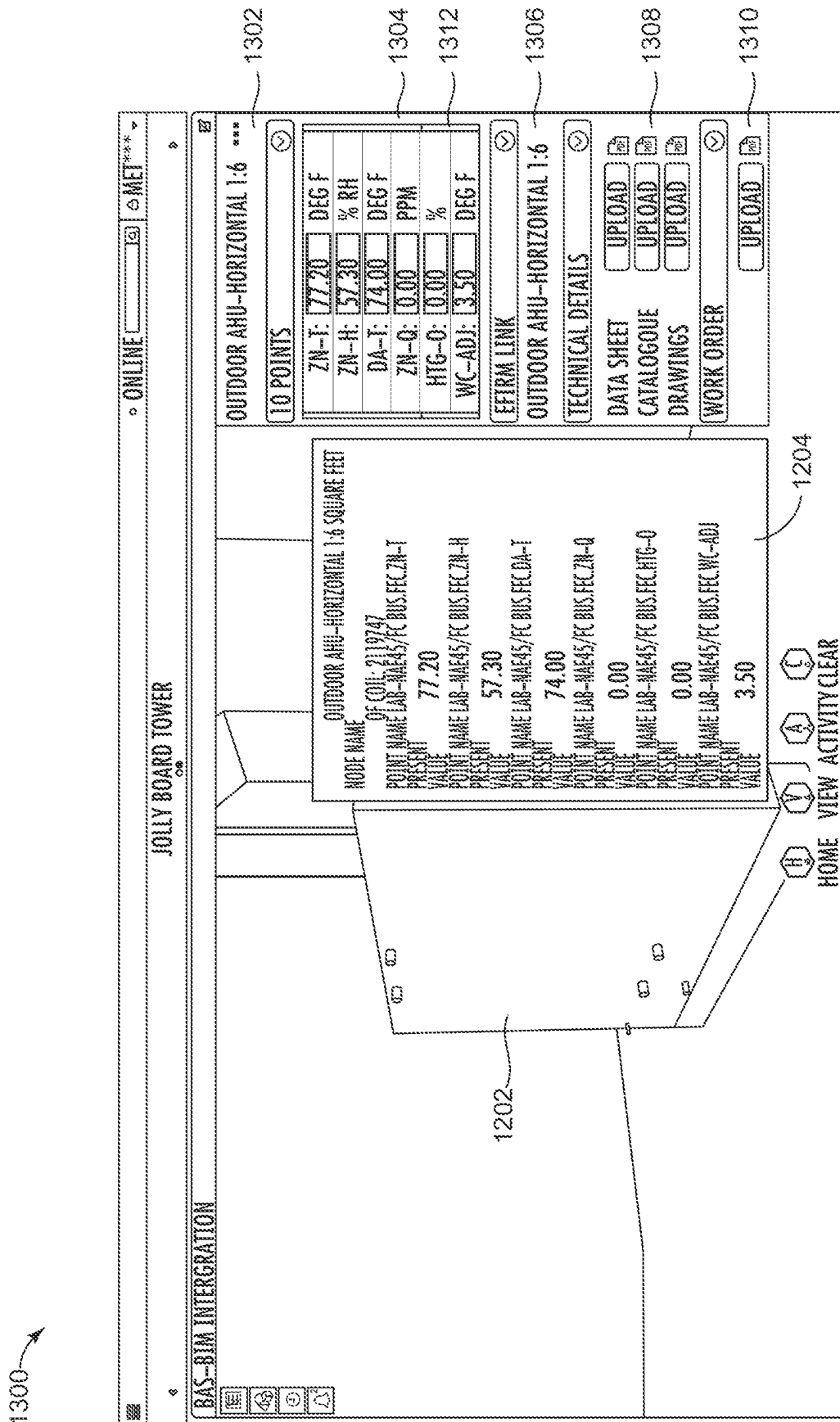


FIG. 12







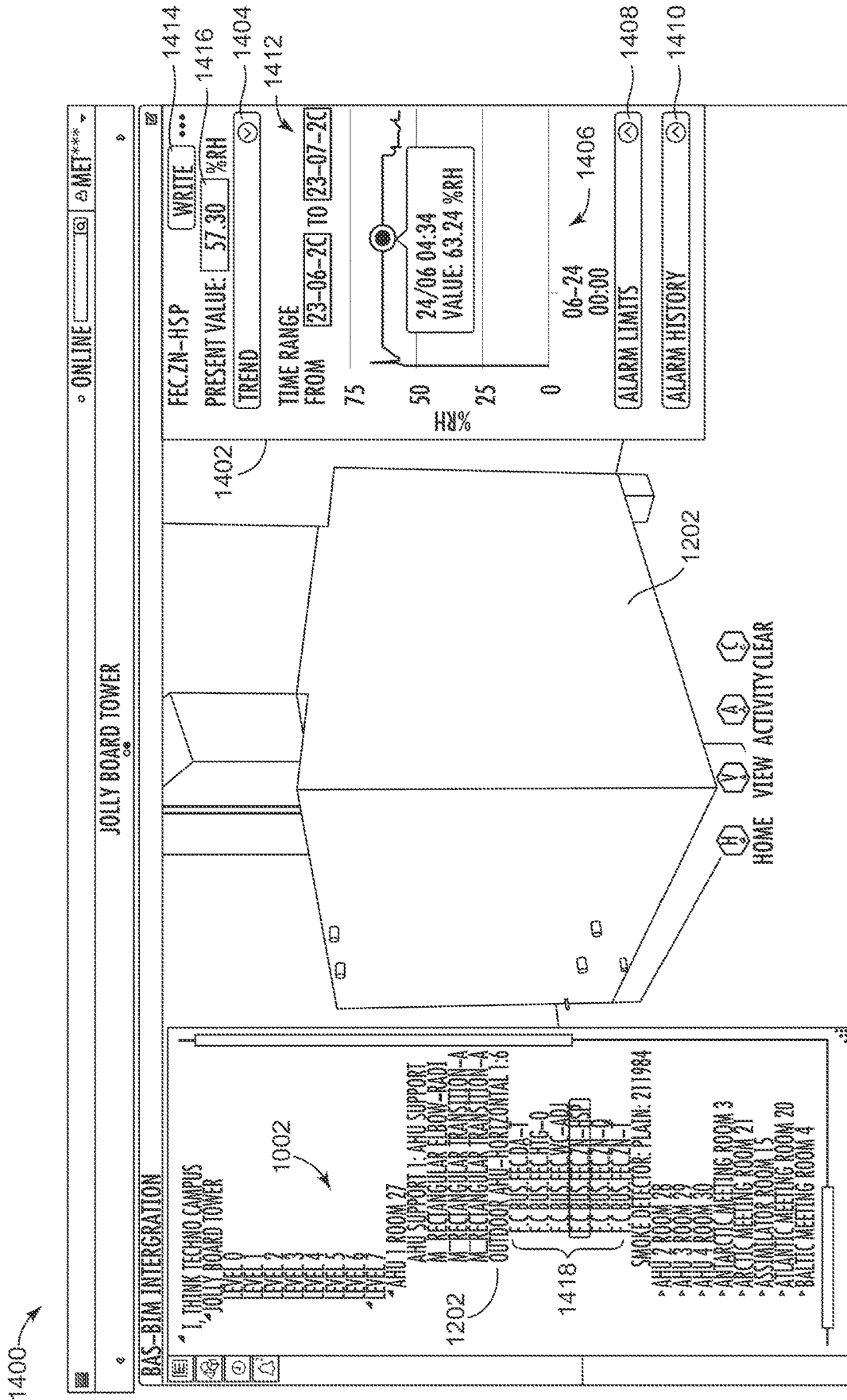


FIG. 14

1500

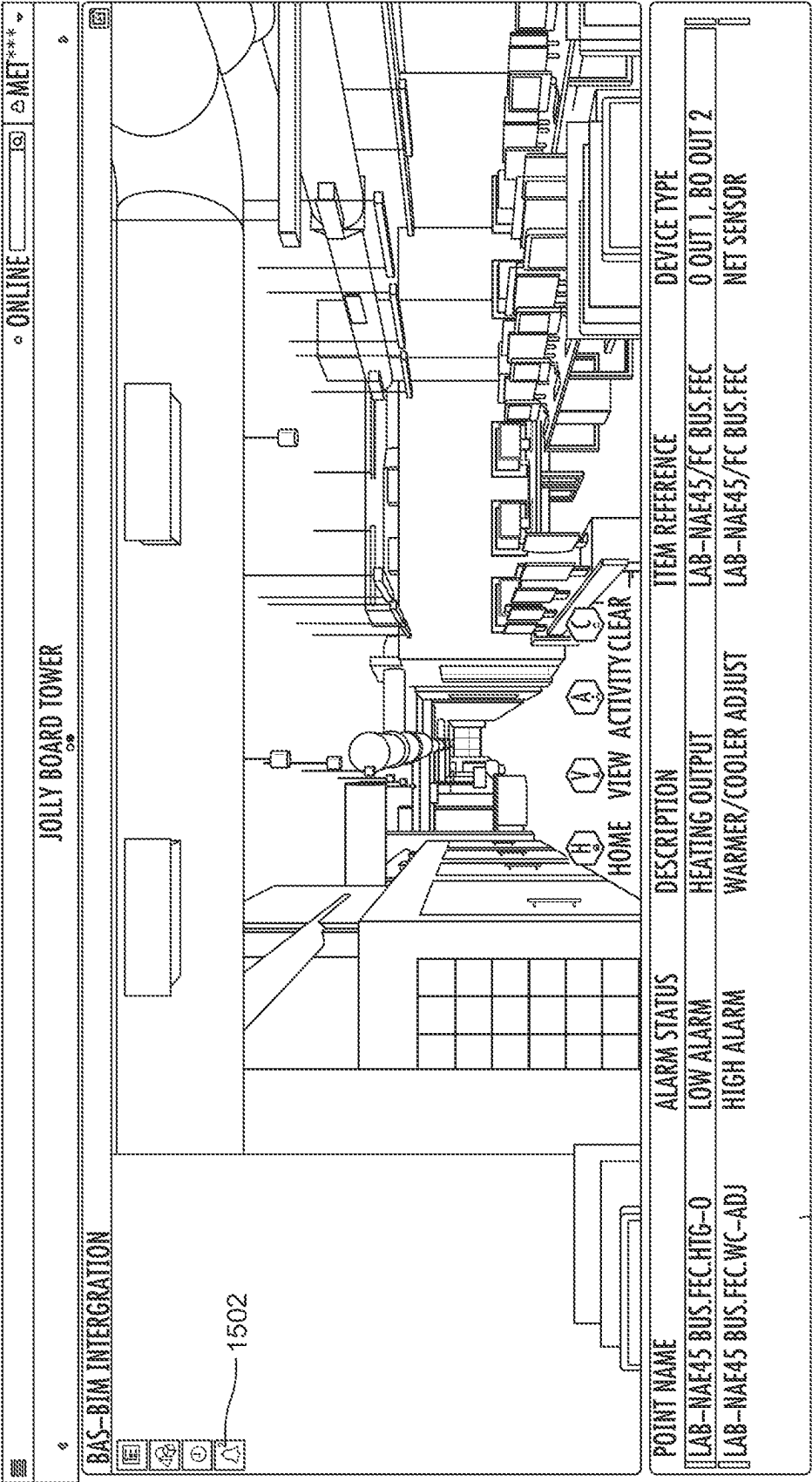


FIG. 15

1504

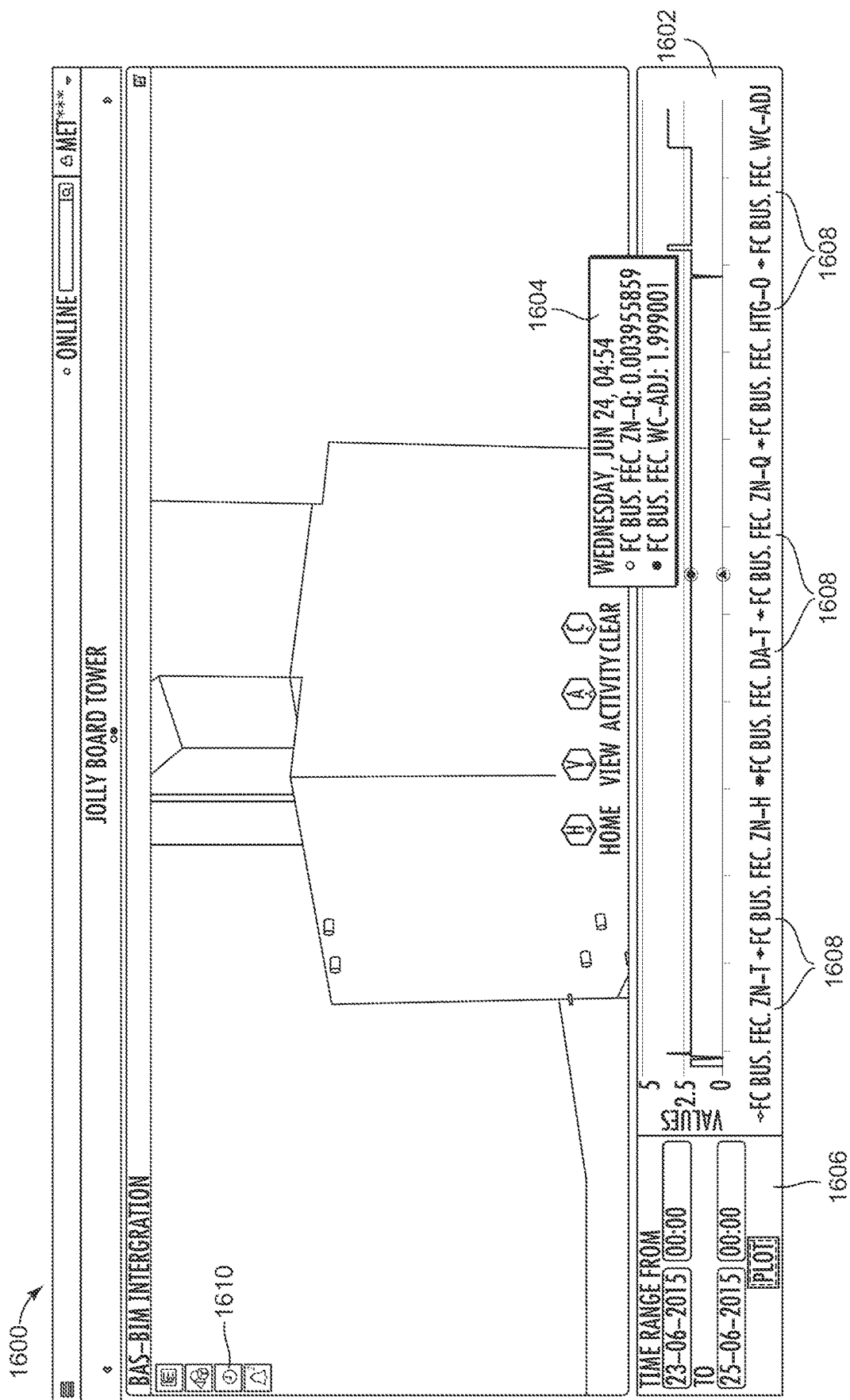


FIG. 16

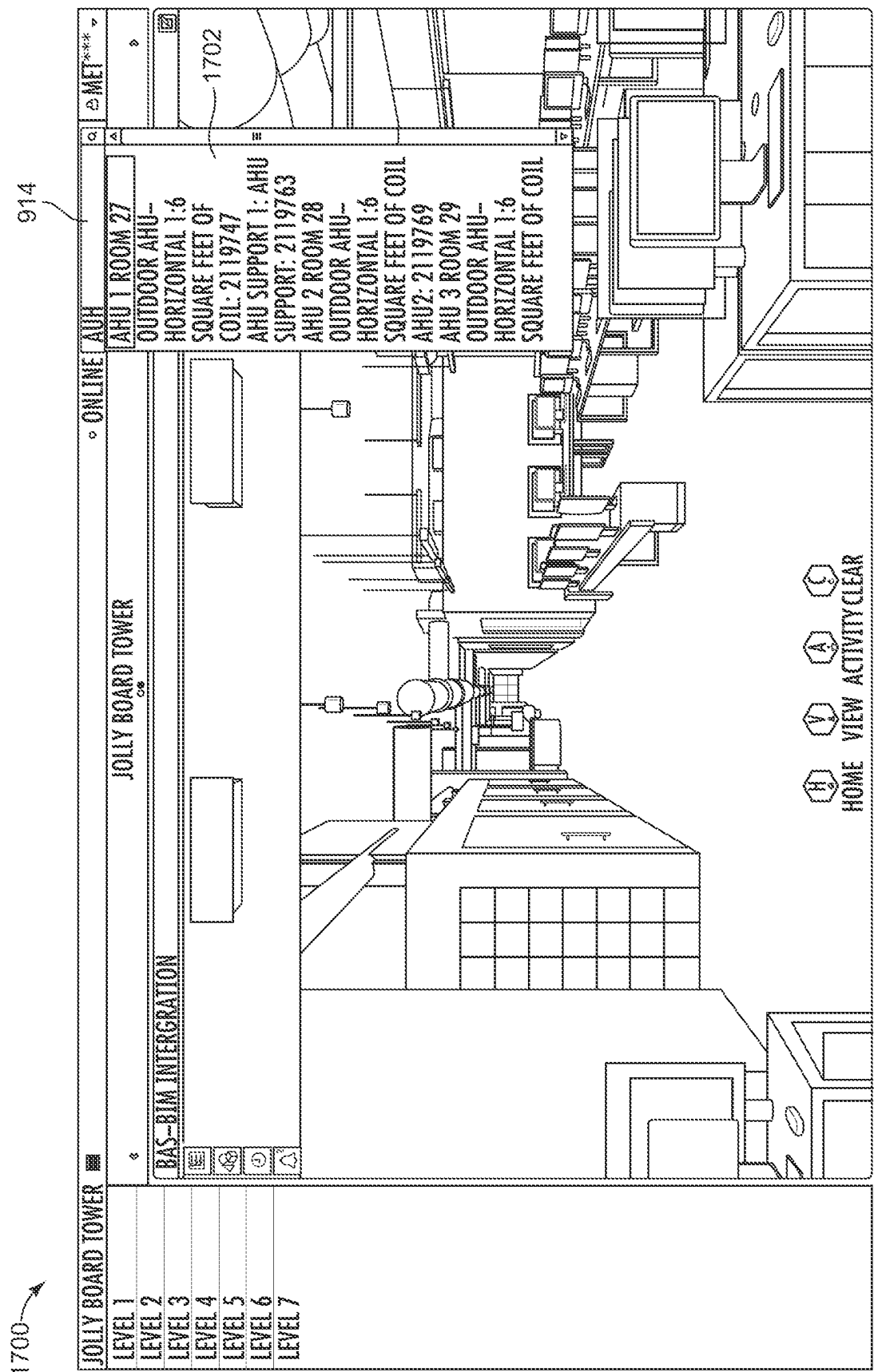


FIG. 17

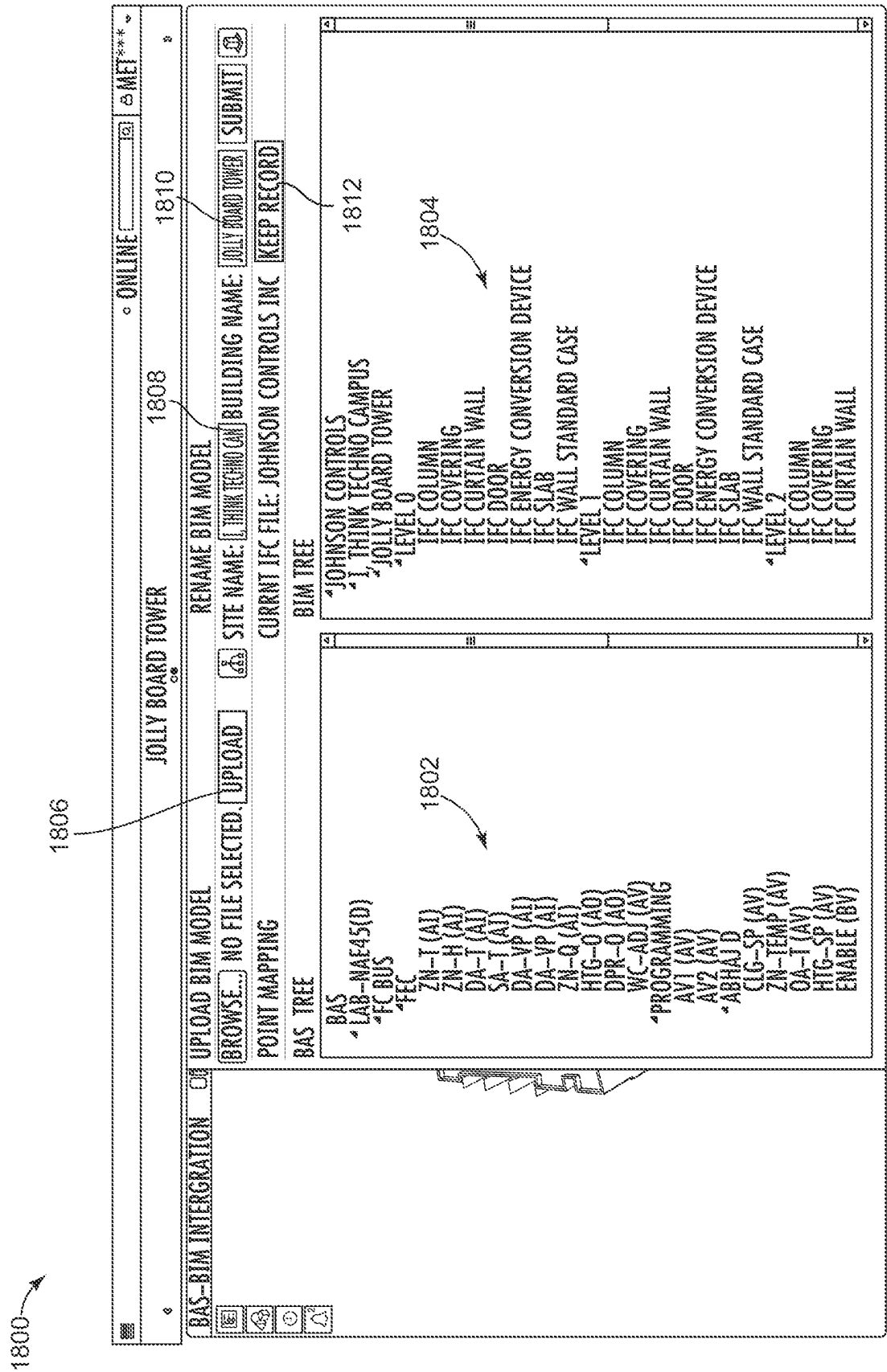


FIG. 18

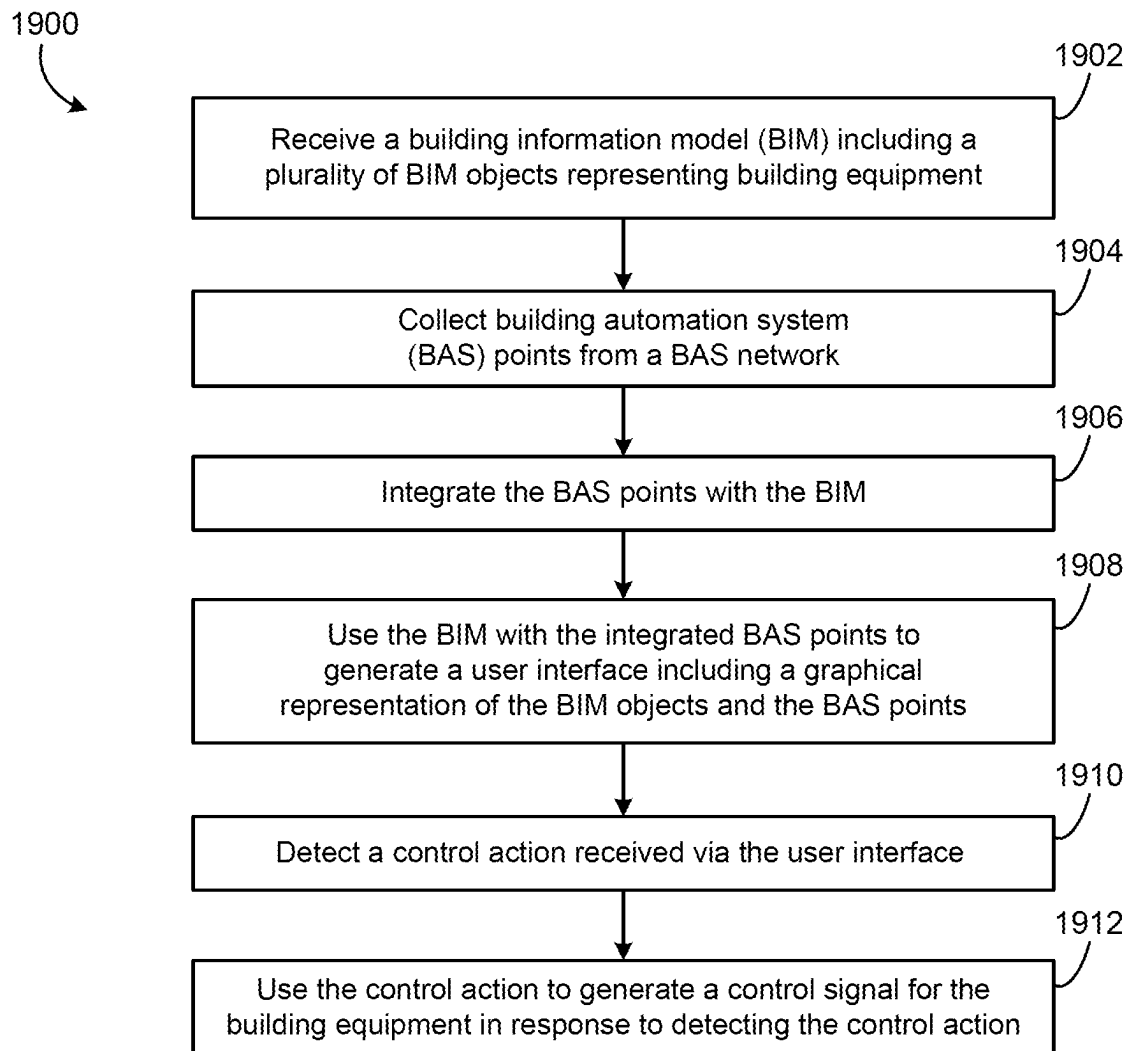


FIG. 19

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BUILDING AUTOMATION SYSTEM WITH INTEGRATED BUILDING INFORMATION MODEL

CROSS-REFERENCE TO RELATED PATENT APPLICATION

This application is a continuation of U.S. patent application Ser. No. 16/694,859 filed Nov. 25, 2019 which is a continuation of U.S. patent application Ser. No. 14/919,516 filed Oct. 21, 2015. The entirety of each of these patent applications is incorporated herein in their entireties.

BACKGROUND

The present invention relates generally to a building automation system (BAS) and more particularly to a BAS configured to integrate BAS data with a building information model (BIM).

A BIM is a representation of the physical and/or functional characteristics of a building. A BIM may represent structural characteristics of the building (e.g., walls, floors, ceilings, doors, windows, etc.) as well as the systems or components contained within the building (e.g., lighting components, electrical systems, mechanical systems, HVAC components, furniture, plumbing systems or fixtures, etc.). In some embodiments, a BIM is a 3D graphical model of the building. A BIM may be created using computer modeling software or other computer-aided design (CAD) tools and may be used by any of a plurality of entities that provide building-related services.

A BAS is, in general, a system of devices configured to control, monitor, and/or manage equipment in or around a building or building area. A BAS can include, for example, a HVAC system, a security system, a lighting system, a fire alerting system, any other system that is capable of managing building functions or devices, or any combination thereof. Some BASs provide graphical user interfaces that allow a user to interact with components of the BAS. Generating graphics for the graphical user interfaces can be time consuming and often results in low quality graphics that do not adequately represent the building equipment. It would be desirable to use the graphics and modeling provided by a BIM as part of the BAS interface. However, it can be difficult and challenging to integrate BAS points with a BIM.

SUMMARY

One implementation of the present disclosure is a building automation system (BAS). The BAS includes building equipment located within a building and a BAS network configured to facilitate communications between the building equipment. The building equipment operate to affect a variable state or condition within the building. The BAS includes a BAS-BIM integrator configured to receive BAS points from the BAS network and to integrate the BAS points with a building information model (BIM). The BIM includes a plurality of BIM objects representing the building equipment. The BAS includes an integrated BAS-BIM viewer configured to use the BIM with the integrated BAS points to generate a user interface. The user interface includes a graphical representation of the BIM objects and the BAS points integrated therewith.

In some embodiments, the BIM includes a three-dimensional model of the building. The BIM objects may include

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one or more objects representing structural components of the building and one or more objects representing spaces within the building.

In some embodiments, the integrated BAS-BIM viewer uses the integrated BAS points to retrieve corresponding point values from the BAS network and displays the point values as part of the user interface. The point values may include at least one of values measured by the building equipment, values generated by the building equipment, setpoints for the building equipment, and operating parameters for the building equipment.

In some embodiments, the integrated BAS-BIM viewer generates a graph including a history of values for at least one of the BAS points and displays the graph as part of the user interface.

In some embodiments, the BAS-BIM integrator includes a BAS tree generator configured to generate a BAS tree comprising the BAS points, a BIM tree generator configured to generate a BIM tree comprising the BIM objects, and a mapping interface generator configured to generate a mapping interface comprising the BAS tree and the BIM tree. The BAS-BIM integrator may be configured to establish mappings between the BAS points and the BIM objects based on a user input received via the mapping interface. In some embodiments, the user input includes dragging and dropping the BAS points from the BAS tree onto BIM objects in the BIM tree.

In some embodiments, the BAS-BIM integrator stores mappings between the BAS points and the BIM objects in a mappings database. The integrated BAS-BIM viewer may retrieve the mappings from the mappings database and use the mappings to generate the user interface.

In some embodiments, the integrated BAS-BIM viewer receives a control action via the user interface and uses the control action to generate a control signal for the building equipment.

Another implementation of the present disclosure is a system for integrating building automation system (BAS) points with a building information model (BIM). The system includes a BAS-BIM integrator configured to receive BAS points from a BAS network and to integrate the BAS points with a BIM. The BIM includes a plurality of BIM objects representing building equipment. The system includes an integrated BAS-BIM viewer configured to use the BIM with the integrated BAS points to generate a user interface. The user interface includes a graphical representation of the BIM objects and the BAS points integrated therewith.

In some embodiments, the BIM includes a three-dimensional model of the building. The BIM objects may include one or more objects representing structural components of the building and one or more objects representing spaces within the building.

In some embodiments, the integrated BAS-BIM viewer uses the integrated BAS points to retrieve corresponding point values from the BAS network and displays the point values as part of the user interface. The point values may include at least one of values measured by the building equipment, values generated by the building equipment, setpoints for the building equipment, and operating parameters for the building equipment.

In some embodiments, the integrated BAS-BIM viewer generates a graph including a history of values for at least one of the BAS points and displays the graph as part of the user interface.

In some embodiments, the BAS-BIM integrator includes a BAS tree generator configured to generate a BAS tree comprising the BAS points, a BIM tree generator configured

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to generate a BIM tree comprising the BIM objects, and a mapping interface generator configured to generate a mapping interface comprising the BAS tree and the BIM tree. The BAS-BIM integrator may be configured to establish mappings between the BAS points and the BIM objects based on a user input received via the mapping interface. In some embodiments, the user input includes dragging and dropping the BAS points from the BAS tree onto BIM objects in the BIM tree.

In some embodiments, the BAS-BIM integrator stores mappings between the BAS points and the BIM objects in a mappings database. The integrated BAS-BIM viewer may retrieve the mappings from the mappings database and use the mappings to generate the user interface.

In some embodiments, the integrated BAS-BIM viewer receives a control action via the user interface and uses the control action to generate a control signal for the building equipment.

Another implementation of the present disclosure is a method for integrating building automation system (BAS) points with a building information model (BIM). The method includes receiving a BIM including a plurality of BIM objects representing building equipment, collecting BAS points from a BAS network, integrating the BAS points with the BIM, and using the BIM with the integrated BAS points to generate a user interface. The user interface includes a graphical representation of the BIM objects and the BAS points integrated therewith. The method includes detecting a control action received via the user interface and using the control action to generate a control signal for the building equipment in response to detecting the control action.

Those skilled in the art will appreciate that the summary is illustrative only and is not intended to be in any way limiting. Other aspects, inventive features, and advantages of the devices and/or processes described herein, as defined solely by the claims, will become apparent in the detailed description set forth herein and taken in conjunction with the accompanying drawings.

BRIEF DESCRIPTION OF THE DRAWINGS

FIG. 1 is a drawing of a building equipped with a building automation system (BAS), according to an exemplary embodiment.

FIG. 2 is a block diagram of a waterside system which may be used to provide heating and/or cooling to the building of FIG. 1, according to an exemplary embodiment.

FIG. 3 is a block diagram of an airside system which may be used to provide heating and/or cooling to the building of FIG. 1, according to an exemplary embodiment.

FIG. 4 is a block diagram of a BAS which may be used to monitor and control building equipment in the building of FIG. 1, according to an exemplary embodiment.

FIG. 5 is a block diagram of a system for integrating BAS data with a building information model (BIM), according to an exemplary embodiment.

FIG. 6 is a block diagram illustrating the BAS-BIM integrator of FIG. 5 in greater detail, according to an exemplary embodiment.

FIG. 7 is a block diagram of another system for integrating BAS data with a BIM, according to an exemplary embodiment.

FIG. 8 is a block diagram of a BAS controller which may be used to integrate BAS data with a BIM, according to an exemplary embodiment.

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FIGS. 9-18 are drawings of user interfaces which may be generated by the systems of FIGS. 5, 7, and/or 8 illustrating a graphical representation of a BIM with integrated BAS data, according to an exemplary embodiment.

FIG. 19 is a flowchart of a process for integrating BAS data with a BIM, according to an exemplary embodiment.

DETAILED DESCRIPTION

Referring generally to the FIGURES, a building automation system (BAS) with an integrated building information model (BIM) is shown, according to an exemplary embodiment. A BAS is, in general, a system of devices configured to control, monitor, and manage equipment in or around a building or building area. A BAS can include, for example, a HVAC system, a security system, a lighting system, a fire alerting system, any other system that is capable of managing building functions or devices, or any combination thereof.

A BIM is a representation of the physical and/or functional characteristics of a building. A BIM may represent structural characteristics of the building (e.g., walls, floors, ceilings, doors, windows, etc.) as well as the systems or components contained within the building (e.g., lighting components, electrical systems, mechanical systems, HVAC components, furniture, plumbing systems or fixtures, etc.). In some embodiments, a BIM is a 3D graphical model of the building. A BIM may be created using computer modeling software or other computer-aided design (CAD) tools and may be used by any of a plurality of entities that provide building-related services.

In some embodiments, a BIM represents building components as objects (e.g., software objects). For example, a BIM may include a plurality of objects that represent physical components within the building as well as building spaces. Each object may include a collection of attributes that define the physical geometry of the object, the type of object, and/or other properties of the object. For example, objects representing building spaces may define the size and location of the building space. Objects representing physical components may define the geometry of the physical component, the type of component (e.g., lighting fixture, air handling unit, wall, etc.), the location of the physical component, a material from which the physical component is constructed, and/or other attributes of the physical component.

The systems and methods described herein may be used to integrate BAS data with a BIM. Advantageously, the integration provided by the present invention allows dynamic BAS data (e.g., data points and their associated values) to be combined with the BIM. The integrated BIM with BAS data can be viewed using an integrated BAS-BIM viewer (e.g., CAD software, a CAD viewer, a web browser, etc.). The BAS-BIM viewer uses the geometric and location information from the BIM to generate 3D representations of physical components and building spaces.

In some embodiments, the BAS-BIM viewer functions as a user interface for monitoring and controlling the various systems and devices represented in the integrated BIM. For example, a user can view real-time data from the BAS and/or trend data for objects represented in the BIM simply by viewing the BIM with integrated BAS data. The user can view BAS points, change the values of BAS points (e.g., setpoints), configure the BAS, and interact with the BAS via the BAS-BIM viewer. These features allow the BIM with integrated BAS data to be used as a building control interface which provides a graphical 3D representation of the

building and the equipment contained therein without requiring a user to manually create or define graphics for various building components. Additional features and advantages of the present invention are described in greater detail below.

Building Automation System and HVAC System

Referring now to FIGS. 1-4, an exemplary building automation system (BAS) and HVAC system in which the systems and methods of the present invention may be implemented are shown, according to an exemplary embodiment. Referring particularly to FIG. 1, a perspective view of a building 10 is shown. Building 10 is served by a BAS which includes a HVAC system 100. HVAC system 100 may include a plurality of HVAC devices (e.g., heaters, chillers, air handling units, pumps, fans, thermal energy storage, etc.) configured to provide heating, cooling, ventilation, or other services for building 10. For example, HVAC system 100 is shown to include a waterside system 120 and an airside system 130. Waterside system 120 may provide a heated or chilled fluid to an air handling unit of airside system 130. Airside system 130 may use the heated or chilled fluid to heat or cool an airflow provided to building 10. An exemplary waterside system and airside system which may be used in HVAC system 100 are described in greater detail with reference to FIGS. 2-3.

HVAC system 100 is shown to include a chiller 102, a boiler 104, and a rooftop air handling unit (AHU) 106. Waterside system 120 may use boiler 104 and chiller 102 to heat or cool a working fluid (e.g., water, glycol, etc.) and may circulate the working fluid to AHU 106. In various embodiments, the HVAC devices of waterside system 120 may be located in or around building 10 (as shown in FIG. 1) or at an offsite location such as a central plant (e.g., a chiller plant, a steam plant, a heat plant, etc.). The working fluid may be heated in boiler 104 or cooled in chiller 102, depending on whether heating or cooling is required in building 10. Boiler 104 may add heat to the circulated fluid, for example, by burning a combustible material (e.g., natural gas) or using an electric heating element. Chiller 102 may place the circulated fluid in a heat exchange relationship with another fluid (e.g., a refrigerant) in a heat exchanger (e.g., an evaporator) to absorb heat from the circulated fluid. The working fluid from chiller 102 and/or boiler 104 may be transported to AHU 106 via piping 108.

AHU 106 may place the working fluid in a heat exchange relationship with an airflow passing through AHU 106 (e.g., via one or more stages of cooling coils and/or heating coils). The airflow may be, for example, outside air, return air from within building 10, or a combination of both. AHU 106 may transfer heat between the airflow and the working fluid to provide heating or cooling for the airflow. For example, AHU 106 may include one or more fans or blowers configured to pass the airflow over or through a heat exchanger containing the working fluid. The working fluid may then return to chiller 102 or boiler 104 via piping 110.

Airside system 130 may deliver the airflow supplied by AHU 106 (i.e., the supply airflow) to building 10 via air supply ducts 112 and may provide return air from building 10 to AHU 106 via air return ducts 114. In some embodiments, airside system 130 includes multiple variable air volume (VAV) units 116. For example, airside system 130 is shown to include a separate VAV unit 116 on each floor or zone of building 10. VAV units 116 may include dampers or other flow control elements that can be operated to control an amount of the supply airflow provided to individual zones of building 10. In other embodiments, airside system 130 delivers the supply airflow into one or more zones of

building 10 (e.g., via supply ducts 112) without using intermediate VAV units 116 or other flow control elements. AHU 106 may include various sensors (e.g., temperature sensors, pressure sensors, etc.) configured to measure attributes of the supply airflow. AHU 106 may receive input from sensors located within AHU 106 and/or within the building zone and may adjust the flow rate, temperature, or other attributes of the supply airflow through AHU 106 to achieve setpoint conditions for the building zone.

Referring now to FIG. 2, a block diagram of a waterside system 200 is shown, according to an exemplary embodiment. In various embodiments, waterside system 200 may supplement or replace waterside system 120 in HVAC system 100 or may be implemented separate from HVAC system 100. When implemented in HVAC system 100, waterside system 200 may include a subset of the HVAC devices in HVAC system 100 (e.g., boiler 104, chiller 102, pumps, valves, etc.) and may operate to supply a heated or chilled fluid to AHU 106. The HVAC devices of waterside system 200 may be located within building 10 (e.g., as components of waterside system 120) or at an offsite location such as a central plant.

In FIG. 2, waterside system 200 is shown as a central plant having a plurality of subplants 202-212. Subplants 202-212 are shown to include a heater subplant 202, a heat recovery chiller subplant 204, a chiller subplant 206, a cooling tower subplant 208, a hot thermal energy storage (TES) subplant 210, and a cold thermal energy storage (TES) subplant 212. Subplants 202-212 consume resources (e.g., water, natural gas, electricity, etc.) from utilities to serve the thermal energy loads (e.g., hot water, cold water, heating, cooling, etc.) of a building or campus. For example, heater subplant 202 may be configured to heat water in a hot water loop 214 that circulates the hot water between heater subplant 202 and building 10. Chiller subplant 206 may be configured to chill water in a cold water loop 216 that circulates the cold water between chiller subplant 206 and building 10. Heat recovery chiller subplant 204 may be configured to transfer heat from cold water loop 216 to hot water loop 214 to provide additional heating for the hot water and additional cooling for the cold water. Condenser water loop 218 may absorb heat from the cold water in chiller subplant 206 and reject the absorbed heat in cooling tower subplant 208 or transfer the absorbed heat to hot water loop 214. Hot TES subplant 210 and cold TES subplant 212 may store hot and cold thermal energy, respectively, for subsequent use.

Hot water loop 214 and cold water loop 216 may deliver the heated and/or chilled water to air handlers located on the rooftop of building 10 (e.g., AHU 106) or to individual floors or zones of building 10 (e.g., VAV units 116). The air handlers push air past heat exchangers (e.g., heating coils or cooling coils) through which the water flows to provide heating or cooling for the air. The heated or cooled air may be delivered to individual zones of building 10 to serve the thermal energy loads of building 10. The water then returns to subplants 202-212 to receive further heating or cooling.

Although subplants 202-212 are shown and described as heating and cooling water for circulation to a building, it is understood that any other type of working fluid (e.g., glycol, CO₂, etc.) may be used in place of or in addition to water to serve the thermal energy loads. In other embodiments, subplants 202-212 may provide heating and/or cooling directly to the building or campus without requiring an intermediate heat transfer fluid. These and other variations to waterside system 200 are within the teachings of the present invention.

Each of subplants **202-212** may include a variety of equipment configured to facilitate the functions of the subplant. For example, heater subplant **202** is shown to include a plurality of heating elements **220** (e.g., boilers, electric heaters, etc.) configured to add heat to the hot water in hot water loop **214**. Heater subplant **202** is also shown to include several pumps **222** and **224** configured to circulate the hot water in hot water loop **214** and to control the flow rate of the hot water through individual heating elements **220**. Chiller subplant **206** is shown to include a plurality of chillers **232** configured to remove heat from the cold water in cold water loop **216**. Chiller subplant **206** is also shown to include several pumps **234** and **236** configured to circulate the cold water in cold water loop **216** and to control the flow rate of the cold water through individual chillers **232**.

Heat recovery chiller subplant **204** is shown to include a plurality of heat recovery heat exchangers **226** (e.g., refrigeration circuits) configured to transfer heat from cold water loop **216** to hot water loop **214**. Heat recovery chiller subplant **204** is also shown to include several pumps **228** and **230** configured to circulate the hot water and/or cold water through heat recovery heat exchangers **226** and to control the flow rate of the water through individual heat recovery heat exchangers **226**. Cooling tower subplant **208** is shown to include a plurality of cooling towers **238** configured to remove heat from the condenser water in condenser water loop **218**. Cooling tower subplant **208** is also shown to include several pumps **240** configured to circulate the condenser water in condenser water loop **218** and to control the flow rate of the condenser water through individual cooling towers **238**.

Hot TES subplant **210** is shown to include a hot TES tank **242** configured to store the hot water for later use. Hot TES subplant **210** may also include one or more pumps or valves configured to control the flow rate of the hot water into or out of hot TES tank **242**. Cold TES subplant **212** is shown to include cold TES tanks **244** configured to store the cold water for later use. Cold TES subplant **212** may also include one or more pumps or valves configured to control the flow rate of the cold water into or out of cold TES tanks **244**.

In some embodiments, one or more of the pumps in waterside system **200** (e.g., pumps **222**, **224**, **228**, **230**, **234**, **236**, and/or **240**) or pipelines in waterside system **200** include an isolation valve associated therewith. Isolation valves may be integrated with the pumps or positioned upstream or downstream of the pumps to control the fluid flows in waterside system **200**. In various embodiments, waterside system **200** may include more, fewer, or different types of devices and/or subplants based on the particular configuration of waterside system **200** and the types of loads served by waterside system **200**.

Referring now to FIG. 3, a block diagram of an airside system **300** is shown, according to an exemplary embodiment. In various embodiments, airside system **300** may supplement or replace airside system **130** in HVAC system **100** or may be implemented separate from HVAC system **100**. When implemented in HVAC system **100**, airside system **300** may include a subset of the HVAC devices in HVAC system **100** (e.g., AHU **106**, VAV units **116**, ducts **112-114**, fans, dampers, etc.) and may be located in or around building **10**. Airside system **300** may operate to heat or cool an airflow provided to building **10** using a heated or chilled fluid provided by waterside system **200**.

In FIG. 3, airside system **300** is shown to include an economizer-type air handling unit (AHU) **302**. Economizer-type AHUs vary the amount of outside air and return air used by the air handling unit for heating or cooling. For example,

AHU **302** may receive return air **304** from building zone **306** via return air duct **308** and may deliver supply air **310** to building zone **306** via supply air duct **312**. In some embodiments, AHU **302** is a rooftop unit located on the roof of building **10** (e.g., AHU **106** as shown in FIG. 1) or otherwise positioned to receive both return air **304** and outside air **314**. AHU **302** may be configured to operate exhaust air damper **316**, mixing damper **318**, and outside air damper **320** to control an amount of outside air **314** and return air **304** that combine to form supply air **310**. Any return air **304** that does not pass through mixing damper **318** may be exhausted from AHU **302** through exhaust damper **316** as exhaust air **322**.

Each of dampers **316-320** may be operated by an actuator. For example, exhaust air damper **316** may be operated by actuator **324**, mixing damper **318** may be operated by actuator **326**, and outside air damper **320** may be operated by actuator **328**. Actuators **324-328** may communicate with an AHU controller **330** via a communications link **332**. Actuators **324-328** may receive control signals from AHU controller **330** and may provide feedback signals to AHU controller **330**. Feedback signals may include, for example, an indication of a current actuator or damper position, an amount of torque or force exerted by the actuator, diagnostic information (e.g., results of diagnostic tests performed by actuators **324-328**), status information, commissioning information, configuration settings, calibration data, and/or other types of information or data that may be collected, stored, or used by actuators **324-328**. AHU controller **330** may be an economizer controller configured to use one or more control algorithms (e.g., state-based algorithms, extremum seeking control (ESC) algorithms, proportional-integral (PI) control algorithms, proportional-integral-derivative (PID) control algorithms, model predictive control (MPC) algorithms, feedback control algorithms, etc.) to control actuators **324-328**.

Still referring to FIG. 3, AHU **302** is shown to include a cooling coil **334**, a heating coil **336**, and a fan **338** positioned within supply air duct **312**. Fan **338** may be configured to force supply air **310** through cooling coil **334** and/or heating coil **336** and provide supply air **310** to building zone **306**. AHU controller **330** may communicate with fan **338** via communications link **340** to control a flow rate of supply air **310**. In some embodiments, AHU controller **330** controls an amount of heating or cooling applied to supply air **310** by modulating a speed of fan **338**.

Cooling coil **334** may receive a chilled fluid from waterside system **200** (e.g., from cold water loop **216**) via piping **342** and may return the chilled fluid to waterside system **200** via piping **344**. Valve **346** may be positioned along piping **342** or piping **344** to control a flow rate of the chilled fluid through cooling coil **334**. In some embodiments, cooling coil **334** includes multiple stages of cooling coils that can be independently activated and deactivated (e.g., by AHU controller **330**, by BAS controller **366**, etc.) to modulate an amount of cooling applied to supply air **310**.

Heating coil **336** may receive a heated fluid from waterside system **200** (e.g., from hot water loop **214**) via piping **348** and may return the heated fluid to waterside system **200** via piping **350**. Valve **352** may be positioned along piping **348** or piping **350** to control a flow rate of the heated fluid through heating coil **336**. In some embodiments, heating coil **336** includes multiple stages of heating coils that can be independently activated and deactivated (e.g., by AHU controller **330**, by BAS controller **366**, etc.) to modulate an amount of heating applied to supply air **310**.

Each of valves **346** and **352** may be controlled by an actuator. For example, valve **346** may be controlled by

actuator 354 and valve 352 may be controlled by actuator 356. Actuators 354-356 may communicate with AHU controller 330 via communications links 358-360. Actuators 354-356 may receive control signals from AHU controller 330 and may provide feedback signals to controller 330. In some embodiments, AHU controller 330 receives a measurement of the supply air temperature from a temperature sensor 362 positioned in supply air duct 312 (e.g., downstream of cooling coil 334 and/or heating coil 336). AHU controller 330 may also receive a measurement of the temperature of building zone 306 from a temperature sensor 364 located in building zone 306.

In some embodiments, AHU controller 330 operates valves 346 and 352 via actuators 354-356 to modulate an amount of heating or cooling provided to supply air 310 (e.g., to achieve a setpoint temperature for supply air 310 or to maintain the temperature of supply air 310 within a setpoint temperature range). The positions of valves 346 and 352 affect the amount of heating or cooling provided to supply air 310 by cooling coil 334 or heating coil 336 and may correlate with the amount of energy consumed to achieve a desired supply air temperature. AHU controller 330 may control the temperature of supply air 310 and/or building zone 306 by activating or deactivating coils 334-336, adjusting a speed of fan 338, or a combination of both.

Still referring to FIG. 3, airside system 300 is shown to include a building automation system (BAS) controller 366 and a client device 368. BAS controller 366 may include one or more computer systems (e.g., servers, supervisory controllers, subsystem controllers, etc.) that serve as system level controllers, application or data servers, head nodes, or master controllers for airside system 300, waterside system 200, HVAC system 100, and/or other controllable systems that serve building 10. BAS controller 366 may communicate with multiple downstream building systems or subsystems (e.g., HVAC system 100, a security system, a lighting system, waterside system 200, etc.) via a communications link 370 according to like or disparate protocols (e.g., LON, BACnet, etc.). In various embodiments, AHU controller 330 and BAS controller 366 may be separate (as shown in FIG. 3) or integrated. In an integrated implementation, AHU controller 330 may be a software module configured for execution by a processor of BAS controller 366.

In some embodiments, AHU controller 330 receives information from BAS controller 366 (e.g., commands, setpoints, operating boundaries, etc.) and provides information to BAS controller 366 (e.g., temperature measurements, valve or actuator positions, operating statuses, diagnostics, etc.). For example, AHU controller 330 may provide BAS controller 366 with temperature measurements from temperature sensors 362-364, equipment on/off states, equipment operating capacities, and/or any other information that can be used by BAS controller 366 to monitor or control a variable state or condition within building zone 306.

Client device 368 may include one or more human-machine interfaces or client interfaces (e.g., graphical user interfaces, reporting interfaces, text-based computer interfaces, client-facing web services, web servers that provide pages to web clients, etc.) for controlling, viewing, or otherwise interacting with HVAC system 100, its subsystems, and/or devices. Client device 368 may be a computer workstation, a client terminal, a remote or local interface, or any other type of user interface device. Client device 368 may be a stationary terminal or a mobile device. For example, client device 368 may be a desktop computer, a computer server with a user interface, a laptop computer, a tablet, a smartphone, a PDA, or any other type of mobile or

non-mobile device. Client device 368 may communicate with BAS controller 366 and/or AHU controller 330 via communications link 372.

Referring now to FIG. 4, a block diagram of a building automation system (BAS) 400 is shown, according to an exemplary embodiment. BAS 400 may be implemented in building 10 to automatically monitor and control various building functions. BAS 400 is shown to include BAS controller 366 and a plurality of building subsystems 428. Building subsystems 428 are shown to include a building electrical subsystem 434, an information communication technology (ICT) subsystem 436, a security subsystem 438, a HVAC subsystem 440, a lighting subsystem 442, a lift/escalators subsystem 432, and a fire safety subsystem 430. In various embodiments, building subsystems 428 can include fewer, additional, or alternative subsystems. For example, building subsystems 428 may also or alternatively include a refrigeration subsystem, an advertising or signage subsystem, a cooking subsystem, a vending subsystem, a printer or copy service subsystem, or any other type of building subsystem that uses controllable equipment and/or sensors to monitor or control building 10. In some embodiments, building subsystems 428 include waterside system 200 and/or airside system 300, as described with reference to FIGS. 2-3.

Each of building subsystems 428 may include any number of devices, controllers, and connections for completing its individual functions and control activities. HVAC subsystem 440 may include many of the same components as HVAC system 100, as described with reference to FIGS. 1-3. For example, HVAC subsystem 440 may include a chiller, a boiler, any number of air handling units, economizers, field controllers, supervisory controllers, actuators, temperature sensors, and other devices for controlling the temperature, humidity, airflow, or other variable conditions within building 10. Lighting subsystem 442 may include any number of light fixtures, ballasts, lighting sensors, dimmers, or other devices configured to controllably adjust the amount of light provided to a building space. Security subsystem 438 may include occupancy sensors, video surveillance cameras, digital video recorders, video processing servers, intrusion detection devices, access control devices and servers, or other security-related devices.

Still referring to FIG. 4, BAS controller 366 is shown to include a communications interface 407 and a BAS interface 409. Interface 407 may facilitate communications between BAS controller 366 and external applications (e.g., monitoring and reporting applications 422, enterprise control applications 426, remote systems and applications 444, applications residing on client devices 448, etc.) for allowing user control, monitoring, and adjustment to BAS controller 366 and/or subsystems 428. Interface 407 may also facilitate communications between BAS controller 366 and client devices 448. BAS interface 409 may facilitate communications between BAS controller 366 and building subsystems 428 (e.g., HVAC, lighting security, lifts, power distribution, business, etc.).

Interfaces 407, 409 can be or include wired or wireless communications interfaces (e.g., jacks, antennas, transmitters, receivers, transceivers, wire terminals, etc.) for conducting data communications with building subsystems 428 or other external systems or devices. In various embodiments, communications via interfaces 407, 409 may be direct (e.g., local wired or wireless communications) or via a communications network 446 (e.g., a WAN, the Internet, a cellular network, etc.). For example, interfaces 407, 409 can include an Ethernet card and port for sending and

receiving data via an Ethernet-based communications link or network. In another example, interfaces 407, 409 can include a WiFi transceiver for communicating via a wireless communications network. In another example, one or both of interfaces 407, 409 may include cellular or mobile phone communications transceivers. In one embodiment, communications interface 407 is a power line communications interface and BAS interface 409 is an Ethernet interface. In other embodiments, both communications interface 407 and BAS interface 409 are Ethernet interfaces or are the same Ethernet interface.

Still referring to FIG. 4, BAS controller 366 is shown to include a processing circuit 404 including a processor 406 and memory 408. Processing circuit 404 may be communicably connected to BAS interface 409 and/or communications interface 407 such that processing circuit 404 and the various components thereof can send and receive data via interfaces 407, 409. Processor 406 can be implemented as a general purpose processor, an application specific integrated circuit (ASIC), one or more field programmable gate arrays (FPGAs), a group of processing components, or other suitable electronic processing components.

Memory 408 (e.g., memory, memory unit, storage device, etc.) may include one or more devices (e.g., RAM, ROM, Flash memory, hard disk storage, etc.) for storing data and/or computer code for completing or facilitating the various processes, layers and modules described in the present application. Memory 408 may be or include volatile memory or non-volatile memory. Memory 408 may include database components, object code components, script components, or any other type of information structure for supporting the various activities and information structures described in the present application. According to an exemplary embodiment, memory 408 is communicably connected to processor 406 via processing circuit 404 and includes computer code for executing (e.g., by processing circuit 404 and/or processor 406) one or more processes described herein.

In some embodiments, BAS controller 366 is implemented within a single computer (e.g., one server, one housing, etc.). In various other embodiments BAS controller 366 may be distributed across multiple servers or computers (e.g., that can exist in distributed locations). Further, while FIG. 4 shows applications 422 and 426 as existing outside of BAS controller 366, in some embodiments, applications 422 and 426 may be hosted within BAS controller 366 (e.g., within memory 408).

Still referring to FIG. 4, memory 408 is shown to include an enterprise integration layer 410, an automated measurement and validation (AM&V) layer 412, a demand response (DR) layer 414, a fault detection and diagnostics (FDD) layer 416, an integrated control layer 418, and a building subsystem integration layer 420. Layers 410-420 may be configured to receive inputs from building subsystems 428 and other data sources, determine optimal control actions for building subsystems 428 based on the inputs, generate control signals based on the optimal control actions, and provide the generated control signals to building subsystems 428. The following paragraphs describe some of the general functions performed by each of layers 410-420 in BAS 400.

Enterprise integration layer 410 may be configured to serve clients or local applications with information and services to support a variety of enterprise-level applications. For example, enterprise control applications 426 may be configured to provide subsystem-spanning control to a graphical user interface (GUI) or to any number of enterprise-level business applications (e.g., accounting systems,

user identification systems, etc.). Enterprise control applications 426 may also or alternatively be configured to provide configuration GUIs for configuring BAS controller 366. In yet other embodiments, enterprise control applications 426 can work with layers 410-420 to optimize building performance (e.g., efficiency, energy use, comfort, or safety) based on inputs received at interface 407 and/or BAS interface 409.

Building subsystem integration layer 420 may be configured to manage communications between BAS controller 366 and building subsystems 428. For example, building subsystem integration layer 420 may receive sensor data and input signals from building subsystems 428 and provide output data and control signals to building subsystems 428. Building subsystem integration layer 420 may also be configured to manage communications between building subsystems 428. Building subsystem integration layer 420 translate communications (e.g., sensor data, input signals, output signals, etc.) across a plurality of multi-vendor/multi-protocol systems.

Demand response layer 414 may be configured to optimize resource usage (e.g., electricity use, natural gas use, water use, etc.) and/or the monetary cost of such resource usage in response to satisfy the demand of building 10. The optimization may be based on time-of-use prices, curtailment signals, energy availability, or other data received from utility providers, distributed energy generation systems 424, from energy storage 427 (e.g., hot TES 242, cold TES 244, etc.), or from other sources. Demand response layer 414 may receive inputs from other layers of BAS controller 366 (e.g., building subsystem integration layer 420, integrated control layer 418, etc.). The inputs received from other layers may include environmental or sensor inputs such as temperature, carbon dioxide levels, relative humidity levels, air quality sensor outputs, occupancy sensor outputs, room schedules, and the like. The inputs may also include inputs such as electrical use (e.g., expressed in kWh), thermal load measurements, pricing information, projected pricing, smoothed pricing, curtailment signals from utilities, and the like.

According to an exemplary embodiment, demand response layer 414 includes control logic for responding to the data and signals it receives. These responses can include communicating with the control algorithms in integrated control layer 418, changing control strategies, changing setpoints, or activating/deactivating building equipment or subsystems in a controlled manner. Demand response layer 414 may also include control logic configured to determine when to utilize stored energy. For example, demand response layer 414 may determine to begin using energy from energy storage 427 just prior to the beginning of a peak use hour.

In some embodiments, demand response layer 414 includes a control module configured to actively initiate control actions (e.g., automatically changing setpoints) which minimize energy costs based on one or more inputs representative of or based on demand (e.g., price, a curtailment signal, a demand level, etc.). In some embodiments, demand response layer 414 uses equipment models to determine an optimal set of control actions. The equipment models may include, for example, thermodynamic models describing the inputs, outputs, and/or functions performed by various sets of building equipment. Equipment models may represent collections of building equipment (e.g., subplants, chiller arrays, etc.) or individual devices (e.g., individual chillers, heaters, pumps, etc.).

Demand response layer 414 may further include or draw upon one or more demand response policy definitions (e.g.,

databases, XML, files, etc.). The policy definitions may be edited or adjusted by a user (e.g., via a graphical user interface) so that the control actions initiated in response to demand inputs may be tailored for the user's application, desired comfort level, particular building equipment, or based on other concerns. For example, the demand response policy definitions can specify which equipment may be turned on or off in response to particular demand inputs, how long a system or piece of equipment should be turned off, what setpoints can be changed, what the allowable set point adjustment range is, how long to hold a high demand setpoint before returning to a normally scheduled setpoint, how close to approach capacity limits, which equipment modes to utilize, the energy transfer rates (e.g., the maximum rate, an alarm rate, other rate boundary information, etc.) into and out of energy storage devices (e.g., thermal storage tanks, battery banks, etc.), and when to dispatch on-site generation of energy (e.g., via fuel cells, a motor generator set, etc.).

Integrated control layer **418** may be configured to use the data input or output of building subsystem integration layer **420** and/or demand response layer **414** to make control decisions. Due to the subsystem integration provided by building subsystem integration layer **420**, integrated control layer **418** can integrate control activities of the subsystems **428** such that the subsystems **428** behave as a single integrated supersystem. In an exemplary embodiment, integrated control layer **418** includes control logic that uses inputs and outputs from a plurality of building subsystems to provide greater comfort and energy savings relative to the comfort and energy savings that separate subsystems could provide alone. For example, integrated control layer **418** may be configured to use an input from a first subsystem to make an energy-saving control decision for a second subsystem. Results of these decisions can be communicated back to building subsystem integration layer **420**.

Integrated control layer **418** is shown to be logically below demand response layer **414**. Integrated control layer **418** may be configured to enhance the effectiveness of demand response layer **414** by enabling building subsystems **428** and their respective control loops to be controlled in coordination with demand response layer **414**. This configuration may advantageously reduce disruptive demand response behavior relative to conventional systems. For example, integrated control layer **418** may be configured to assure that a demand response-driven upward adjustment to the setpoint for chilled water temperature (or another component that directly or indirectly affects temperature) does not result in an increase in fan energy (or other energy used to cool a space) that would result in greater total building energy use than was saved at the chiller.

Integrated control layer **418** may be configured to provide feedback to demand response layer **414** so that demand response layer **414** checks that constraints (e.g., temperature, lighting levels, etc.) are properly maintained even while demanded load shedding is in progress. The constraints may also include setpoint or sensed boundaries relating to safety, equipment operating limits and performance, comfort, fire codes, electrical codes, energy codes, and the like. Integrated control layer **418** is also logically below fault detection and diagnostics layer **416** and automated measurement and validation layer **412**. Integrated control layer **418** may be configured to provide calculated inputs (e.g., aggregations) to these higher levels based on outputs from more than one building subsystem.

Automated measurement and validation (AM&V) layer **412** may be configured to verify that control strategies

commanded by integrated control layer **418** or demand response layer **414** are working properly (e.g., using data aggregated by AM&V layer **412**, integrated control layer **418**, building subsystem integration layer **420**, FDD layer **416**, or otherwise). The calculations made by AM&V layer **412** may be based on building system energy models and/or equipment models for individual BAS devices or subsystems. For example, AM&V layer **412** may compare a model-predicted output with an actual output from building subsystems **428** to determine an accuracy of the model.

Fault detection and diagnostics (FDD) layer **416** may be configured to provide on-going fault detection for building subsystems **428**, building subsystem devices (i.e., building equipment), and control algorithms used by demand response layer **414** and integrated control layer **418**. FDD layer **416** may receive data inputs from integrated control layer **418**, directly from one or more building subsystems or devices, or from another data source. FDD layer **416** may automatically diagnose and respond to detected faults. The responses to detected or diagnosed faults may include providing an alert message to a user, a maintenance scheduling system, or a control algorithm configured to attempt to repair the fault or to work-around the fault.

FDD layer **416** may be configured to output a specific identification of the faulty component or cause of the fault (e.g., loose damper linkage) using detailed subsystem inputs available at building subsystem integration layer **420**. In other exemplary embodiments, FDD layer **416** is configured to provide "fault" events to integrated control layer **418** which executes control strategies and policies in response to the received fault events. According to an exemplary embodiment, FDD layer **416** (or a policy executed by an integrated control engine or business rules engine) may shut-down systems or direct control activities around faulty devices or systems to reduce energy waste, extend equipment life, or assure proper control response.

FDD layer **416** may be configured to store or access a variety of different system data stores (or data points for live data). FDD layer **416** may use some content of the data stores to identify faults at the equipment level (e.g., specific chiller, specific AHU, specific terminal unit, etc.) and other content to identify faults at component or subsystem levels. For example, building subsystems **428** may generate temporal (i.e., time-series) data indicating the performance of BAS **400** and the various components thereof. The data generated by building subsystems **428** may include measured or calculated values that exhibit statistical characteristics and provide information about how the corresponding system or process (e.g., a temperature control process, a flow control process, etc.) is performing in terms of error from its setpoint. These processes can be examined by FDD layer **416** to expose when the system begins to degrade in performance and alert a user to repair the fault before it becomes more severe.

BAS-BIM Integration

Referring now to FIG. 5, a system **500** for integrating building automation system data with a building information model is shown, according to an exemplary embodiment. A building information model (BIM) is a representation of the physical and/or functional characteristics of a building. A BIM may represent structural characteristics of the building (e.g., walls, floors, ceilings, doors, windows, etc.) as well as the systems or components contained within the building (e.g., lighting components, electrical systems, mechanical systems, HVAC components, furniture, plumbing systems or fixtures, etc.).

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In some embodiments, a BIM is a 3D graphical model of the building. A BIM may be created using computer modeling software or other computer-aided design (CAD) tools and may be used by any of a plurality of entities that provide building-related services. For example, a BIM may be used by architects, contractors, landscape architects, surveyors, civil engineers, structural engineers, building services engineers, building owners/operators, or any other entity to obtain information about the building and/or the components contained therein. A BIM may replace 2D technical drawings (e.g., plans, elevations, sections, etc.) and may provide significantly more information than traditional 2D drawings. For example, a BIM may include spatial relationships, light analyses, geographic information, and/or qualities or properties of building components (e.g., manufacturer details).

In some embodiments, a BIM represents building components as objects (e.g., software objects). For example, a BIM may include a plurality of objects that represent physical components within the building as well as building spaces. Each object may include a collection of attributes that define the physical geometry of the object, the type of object, and/or other properties of the object. For example, objects representing building spaces may define the size and location of the building space. Objects representing physical components may define the geometry of the physical component, the type of component (e.g., lighting fixture, air handling unit, wall, etc.), the location of the physical component, a material from which the physical component is constructed, and/or other attributes of the physical component.

In some embodiments, a BIM includes an industry foundation class (IFC) data model that describes building and construction industry data. An IFC data model is an object-based file format that facilitates interoperability in the architecture, engineering, and construction industry. An IFC model may store and represent building components in terms of a data schema. An IFC model may include multiple layers and may include object definitions (e.g., IfcObjectDefinition), relationships (e.g., IfcRelationship), and property definitions (e.g., IfcPropertyDefinition). Object definitions may identify various objects in the IFC model and may include information such as physical placement, controls, and groupings. Relationships may capture relationships between objects such as composition, assignment, connectivity, association, and definition. Property definitions may capture dynamically extensible properties about objects. Any type of property may be defined as an enumeration, a list of values, a table of values, or a data structure.

A BIM can be viewed and manipulated using a 3D modeling program (e.g., CAD software), a model viewer, a web browser, and/or any other software capable of interpreting and rendering the information contained within the BIM. Appropriate viewing software may allow a user to view the representation of the building from any of a variety of perspectives and/or locations. For example, a user can view the BIM from a perspective within the building to see how the building would look from that location. In other words, a user can simulate the perspective of a person within the building.

Advantageously, the integration provided by system 500 allows dynamic BAS data (e.g., data points and their associated values) to be combined with the BIM. The integrated BIM with BAS data can be viewed using an integrated BAS-BIM viewer (e.g., CAD software, a CAD viewer, a web browser, etc.). The BAS-BIM viewer uses the geometric and location information from the BIM to generate 3D

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representations of physical components and building spaces. In some embodiments, the BAS-BIM viewer functions as a user interface for monitoring and controlling the various systems and devices represented in the integrated BIM. For example, a user can view real-time data from the BAS and/or trend data for objects represented in the BIM simply by viewing the BIM with integrated BAS data. The user can view BAS points, change the values of BAS points (e.g., setpoints), configure the BAS, and interact with the BAS via the BAS-BIM viewer. These features allow the BIM with integrated BAS data to be used as a building control interface which provides a graphical 3D representation of the building and the equipment contained therein without requiring a user to manually create or define graphics for various building components.

Still referring to FIG. 5, system 500 is shown to include a BAS-BIM integrator 502, an integrated BAS-BIM viewer 504, a BIM database 506, a user interface 508, a BAS network 510, and building equipment 512. In some embodiments, some or all of the components of system 500 are part of BAS 400. For example, BAS network 510 may be a building automation and control network (e.g., a BACnet network, a LonWorks network, etc.) used by BAS 400 to communicate with building equipment 512. Building equipment 512 may include any of the equipment described with reference to FIGS. 1-4. For example, building equipment 512 may include HVAC equipment (e.g., chillers, boilers, air handling units, pumps, fans, valves, dampers, etc.), fire safety equipment, lifts/escalators, electrical equipment, communications equipment, security equipment, lighting equipment, or any other type of equipment which may be contained within a building.

In some embodiments, BAS-BIM integrator 502, integrated BAS-BIM viewer 504, BIM database 506, and user interface 508 are components of BAS controller 366. In other embodiments, one or more of components 502-508 may be components of a user device. For example, integrated BAS-BIM viewer 504 may be an application running on the user device and may be configured to present a BIM with integrated BAS points via a user interface (e.g., user interface 508) of the user device. BAS-BIM integrator 502 may be part of the same application and may be configured to integrate BAS points with a BIM model based on user input provided via user interface 508. In further embodiments, integrated BAS-BIM viewer 504 is part of a user device that receives a BIM with integrated BAS points from a remote BAS-BIM integrator 502. It is contemplated that components 502-508 may be part of the same system/device (e.g., BAS controller 366, a user device, etc.) or may be distributed across multiple systems/devices. All such embodiments are within the scope of the present disclosure.

Still referring to FIG. 5, BAS-BIM integrator 502 is shown receiving a BIM and BAS points. In some embodiments, BAS-BIM integrator 502 receives a BIM from BIM database 506. In other embodiments, the BIM is uploaded by a user or retrieved from another data source. BAS-BIM integrator 502 may receive BAS points from BAS network 510 (e.g., a BACnet network, a LonWorks network, etc.). The BAS points may be measured data points, calculated data points, setpoints, or other types of data points used by the BAS, generated by the BAS, or stored within the BAS (e.g., configuration settings, control parameters, equipment information, alarm information, etc.).

BAS-BIM integrator 502 may be configured to integrate the BAS points with the BIM. In some embodiments, BAS-BIM integrator 502 integrates the BAS points with the BIM based on a user-defined mapping. For example, BAS-

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BIM integrator **502** may be configured to generate a mapping interface presents the BAS points as a BAS tree and presents the BIM objects as a BIM tree. The BAS tree and the BIM tree may be presented to a user via user interface **508**. The mapping interface may allow a user to drag and drop BAS points onto objects of the BIM or otherwise define associations between BAS points and BIM objects. An exemplary mapping interface is described in greater detail with reference to FIG. **18**. In other embodiments, BAS-BIM integrator **502** automatically maps the BAS points to BIM objects based on attributes of the BAS points and the BIM objects (e.g., name, attributes, type, etc.).

In some embodiments, BAS-BIM integrator **502** updates or modifies the BIM to include the BAS points. For example, BAS-BIM integrator **502** may store the BAS points as properties or attributes of objects within the BIM (e.g., objects representing building equipment or spaces). The modified BIM with integrated BAS points may be provided to integrated BAS-BIM viewer **504** and/or stored in BIM database **506**. When the BIM is viewed, the BAS points can be viewed along with the other attributes of the BIM objects. In other embodiments, BAS-BIM integrator **502** generates a mapping between BIM objects and BAS points without modifying the BIM. The mapping may be stored in a separate database or included within the BIM. When the BIM is viewed, integrated BAS-BIM viewer **504** may use the mapping to identify BAS points associated with BIM objects.

Integrated BAS-BIM viewer **504** is shown receiving the BIM with integrated BAS points from BAS-BIM integrator **502**. Integrated BAS-BIM viewer **504** may generate a 3D graphical representation of the building and the components contained therein, according to the attributes of objects defined by the BIM. As previously described, the BIM objects may be modified to include BAS points. For example, some or all of the objects within the BIM may be modified to include an attribute identifying a particular BAS point (e.g., a point name, a point ID, etc.). When integrated BAS-BIM viewer **504** renders the BIM with integrated BAS points, integrated BAS-BIM viewer **504** may use the identities of the BAS points provided by the BIM to retrieve corresponding point values from BAS network **510**. Integrated BAS-BIM viewer **504** may incorporate the BAS point values within the BIM to generate a BIM with integrated BAS points and values.

Integrated BAS-BIM viewer **504** is shown providing the BIM with integrated BAS points and values to user interface **508**. User interface **508** may present the BIM with integrated BAS points and values to a user. Advantageously, the BIM with integrated BAS points and values may include real-time data from BAS network **510**, as defined by the integrated BAS points. A user can monitor the BAS and view present values of the BAS points from within the BIM. In some embodiments, the BIM with integrated BAS points and values includes trend data for various BAS points. User interface **508** may display the trend data to a user along with the BIM.

In some embodiments, integrated BAS-BIM viewer **504** receives control actions via user interface **508**. For example, a user can write new values for any of the BAS points displayed in the BIM (e.g., setpoints), send operating commands or control signals to the building equipment displayed in the BIM, or otherwise interact with the BAS via the BIM. Control actions submitted via user interface **508** may be received at integrated BAS-BIM viewer **504** and provided to BAS network **510**. BAS network **510** may use the control actions to generate control signals for building

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equipment **512** or otherwise adjust the operation of building equipment **512**. In this way, the BIM with integrated BAS points and values not only allows a user to monitor the BAS, but also provides the control functionality of a graphical BAS management and control interface. Several examples of the control interface provided by the BIM with integrated BAS points and values are described in greater detail with reference to FIGS. **9-18**.

Referring now to FIG. **6**, a block diagram illustrating BAS-BIM integrator **502** in greater detail is shown, according to an exemplary embodiment. BAS-BIM integrator **502** is shown to include a BIM tree generator **606** and a BAS tree generator **604**. BIM tree generator **606** may be configured to receive a BIM from BIM database **506**. Alternatively, the BIM may be uploaded by a user or retrieved from another location. BIM tree generator **606** may generate a BIM tree based on the BIM. The BIM tree may include a hierarchical listing of BIM objects referenced in the BIM. BAS tree generator **604** may receive BAS points from the BAS and may generate a BAS tree based on the BAS points. The BAS tree may include a hierarchical listing of BAS points. Exemplary BAS and BIM trees are shown in FIG. **18**.

BAS-BIM integrator **502** is shown to include a mapping interface generator **602**. Mapping interface generator **602** may be configured to generate an interface for mapping BAS points to BIM objects. In some embodiments, the mapping interface includes the BAS tree and BIM tree. For example, the BAS tree may be displayed in a first portion of the mapping interface and the BIM tree may be displayed in a second portion of the mapping interface. The mapping interface may be presented to a user via user interface **508**. A user can define point mappings by dragging and dropping BAS points from the BAS tree onto BIM objects in the BIM tree. Mapping interface generator **602** may receive the point mappings from user interface **508** and may provide the point mappings to BIM updater **608**. An exemplary mapping interface which may be generated by mapping interface generator **602** is shown in FIG. **18**.

BIM updater **608** may be configured to update or modify the BIM based on the BAS point mappings. For example, BIM updater **608** may store the BAS points as properties or attributes of objects within the BIM (e.g., objects representing building equipment or spaces). The modified BIM with integrated BAS points may be provided to integrated BAS-BIM viewer **504** and/or stored in BIM database **506**. When the BIM is viewed, the BAS points mapped to a BIM object can be viewed along with other attributes of the BIM objects.

Referring now to FIG. **7**, another system **700** for integrating building automation system data with a building information model is shown, according to an exemplary embodiment. System **700** is shown to include many of the same components as system **500**. For example, system **700** is shown to include a BAS-BIM integrator **502**, an integrated BAS-BIM viewer **504**, a BIM database **506**, a user interface **508**, a BAS network **510**, and building equipment **512**. These components may be the same or similar as previously described with reference to FIGS. **5-6**.

System **700** is also shown to include a point mappings database **702**. In the embodiment shown in FIG. **7**, BAS-BIM integrator **502** does not modify the BIM to include the BAS points, but rather stores the point mappings in point mappings database **702**. When the BIM is viewed, integrated BAS-BIM viewer **504** may retrieve the BIM from BIM database **506** and may retrieve the point mappings from point mappings database **702**. Integrated BAS-BIM viewer

504 may use the point mappings to identify BAS points associated with the BIM objects.

Integrated BAS-BIM viewer **504** may generate a 3D graphical representation of the building and the components contained therein, according to the attributes of objects defined by the BIM. When integrated BAS-BIM viewer **504** renders the BIM, integrated BAS-BIM viewer **504** may use the identities of the BAS points provided by the point mappings to retrieve corresponding point values from BAS network **510**. Integrated BAS-BIM viewer **504** may incorporate the BAS point values within the BIM to generate a BIM with integrated BAS points and values.

Integrated BAS-BIM viewer **504** is shown providing the BIM with integrated BAS points and values to user interface **508**. User interface **508** may present the BIM with integrated BAS points and values to a user. Advantageously, the BIM with integrated BAS points and values may include real-time data from BAS network **510**, as defined by the integrated BAS points. A user can monitor the BAS and view present values of the BAS points from within the BIM. In some embodiments, the BIM with integrated BAS points and values includes trend data for various BAS points. User interface **508** may display the trend data to a user along with the BIM.

Referring now to FIG. 8, a block diagram of a BAS controller **802** is shown, according to an exemplary embodiment. In some embodiments, many of the components of systems **500-700** are components of BAS controller **802**. For example, BAS controller **802** is shown to include a BAS-BIM integrator **502**, an integrated BAS-BIM viewer **504**, a user interface **508**, a mapping interface generator **602**, a BAS tree generator **604**, a BIM tree generator **606**, and a point mappings database **702**. These components may be the same or similar as previously described with reference to FIGS. 5-7. Controller **802** may also include some or all of the components of BAS controller **366**, as described with reference to FIGS. 3-4.

Controller **802** is shown to include a data communications interface **804** and a processing circuit **808**. Interface **804** may facilitate communications between BAS controller **802** and external systems or applications (e.g., BIM database **506**, BAS network **510**, building equipment **512**, a user device, etc.). Interface **804** may include wired or wireless communications interfaces (e.g., jacks, antennas, transmitters, receivers, transceivers, wire terminals, etc.) for conducting data communications with BIM database **506**, BAS network **510**, or other external systems or devices. In various embodiments, communications via interface **804** may be direct (e.g., local wired or wireless communications) or via a communications network (e.g., a WAN, the Internet, a cellular network, etc.). For example, interface **804** may include an Ethernet card and port for sending and receiving data via an Ethernet-based communications link or network. In another example, interface **804** may include a WiFi transceiver for communicating via a wireless communications network, a cellular or mobile phone communications transceiver, or a power line communications interface.

Processing circuit **808** is shown to include a processor **810** and memory **812**. Processor **810** may be a general purpose or specific purpose processor, an application specific integrated circuit (ASIC), one or more field programmable gate arrays (FPGAs), a group of processing components, or other suitable processing components. Processor **810** is configured to execute computer code or instructions stored in memory **812** or received from other computer readable media (e.g., CDROM, network storage, a remote server, etc.).

Memory **812** may include one or more devices (e.g., memory units, memory devices, storage devices, etc.) for storing data and/or computer code for completing and/or facilitating the various processes described in the present disclosure. Memory **812** may include random access memory (RAM), read-only memory (ROM), hard drive storage, temporary storage, non-volatile memory, flash memory, optical memory, or any other suitable memory for storing software objects and/or computer instructions. Memory **812** may include database components, object code components, script components, or any other type of information structure for supporting the various activities and information structures described in the present disclosure. Memory **812** may be communicably connected to processor **810** via processing circuit **818** and may include computer code for executing (e.g., by processor **810**) one or more processes described herein. When processor **810** executes instructions stored in memory **812**, processor **810** generally configures BAS controller **802** (and more particularly processing circuit **808**) to complete such activities.

Still referring to FIG. 8, memory **812** is shown to include a BIM selector **814**. BIM selector **814** may be configured to receive a selected BIM from user **820**. In some embodiments, user **820** uploads the BIM to BAS controller **802**. In other embodiments, BIM selector **814** retrieves the BIM from BIM database **606**. BIM selector **814** may provide the BIM to BIM tree generator **606** for use in generating the BIM tree. In some embodiments, BIM selector **814** provides the BIM to integrated BAS-BIM viewer **504**. In other embodiments, BIM selector **814** provides the BIM tree to BAS-BIM point integrator **502**.

BAS tree generator **604** may receive the BAS points from BAS network **810** via data communications interface **804** and may use the BAS points to generate a BAS tree. The BIM tree and the BAS tree may be provided to mapping interface generator **602**. Mapping interface generator **602** uses the BAS tree and BIM tree to generate a mapping interface. The mapping interface may be presented to user **820** via user interface **508**. The user interacts with the mapping interface to define point mappings. The point mappings may be stored in point mappings database **702** and/or used by BAS-BIM integrator **502** to modify the BIM.

Integrated BAS-BIM viewer **504** may receive the point mappings from point mappings database and may use the point mappings to identify BAS points associated with BIM objects referenced in the BIM. In other embodiments, integrated BAS-BIM viewer **504** receives a BIM with integrated BAS points from BAS-BIM point integrator **502**, as described with reference to FIG. 5. Integrated BAS-BIM viewer **504** may retrieve corresponding point values from BAS network **510** via data communications interface **804**. Integrated BAS-BIM viewer **504** may then present the BIM with integrated BAS points and values to user **820** via user interface **508**.

Still referring to FIG. 8, memory **812** is shown to include an alarm manager **816** and an equipment controller **818**. Alarm manager **816** may receive alarms from BAS network **510** or may identify alarms based on the values of the BAS points. For example, alarm manager **816** may compare the values of the BAS points to alarm thresholds. If a BAS point is not within a range of values defined by the alarm thresholds, alarm manager **816** may determine that an alarm condition exists for the BAS point. Alarm manager **816** may provide alarms to integrated BAS-BIM viewer **504**. Integrated BAS-BIM viewer **504** may use the alarms to generate part of the user interface provided to user **820**.

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Equipment controller **818** may receive control actions from integrated BAS-BIM viewer **504**. The control actions may be user-defined control actions provided via the integrated BAS-BIM viewing interface. Equipment controller **818** may use the control actions to generate control signals for building equipment **512** or otherwise adjust the operation of building equipment **512**. In this way, the BIM with integrated BAS points and values not only allows a user to monitor the BAS, but also provides the control functionality of a graphical BAS management and control interface. Several exemplary graphical interfaces which may be generated by integrated BAS-BIM viewer **504** are described in greater detail with reference to FIGS. 9-18.

User Interfaces

Referring now to FIGS. 9-18, several user interfaces **900-1800** which may be generated by BAS-BIM integrator **502** and integrated BAS-BIM viewer **504** are shown, according to an exemplary embodiment. In some embodiments, interfaces **900-1800** are web interfaces and may be presented via a web browser running on a user device. The user device may be a computer workstation, a client terminal, a personal computer, or any other type of user device. In various embodiments, the user device may be a mobile device (e.g., a smartphone, a tablet, a PDA, a laptop, etc.) or a non-mobile device. In other embodiments, interfaces **900-1800** are presented via a specialized monitoring and control application. The application may run on BAS controller **366**, on a computer system within BAS **400**, on a server, or on a user device. In some embodiments, the application is a mobile application configured to run on a mobile device.

Referring particularly to FIG. 9, an integrated BAS-BIM viewer interface **900** is shown, according to an exemplary embodiment. Interface **900** may be generated by integrated BAS-BIM viewer **504** to view and interact with a BIM with integrated BAS points and values. Interface **900** is shown to include a perspective view of a building **902**. Building **902** is shown to include walls **904**, a roof **906**, and windows **908**. The geometry and locations of components **904-906** may be defined by the BIM objects within the BIM model.

Interface **900** may be interactive and may allow a user to view building **902** from multiple different angles and/or perspectives. For example, interface **900** may provide interface options for zooming in, zooming out, panning vertically or horizontally, rotating the view, and/or otherwise changing the perspective. View buttons **912** may be used to select a particular view (e.g., top, side, front, back, left, right, back, perspective, etc.) of building **902**. Navigation buttons **910** may be used to display a tree of BIM objects, filter the BIM objects (e.g., by type, by location, etc.), display any alarms provided by the BAS, or otherwise manipulate the view of building **902**. Search box **914** can be used to search for particular BIM objects, search for BAS points, search for a particular room or zone, and/or search for building equipment. Selecting an item via navigation buttons **910** or search box **914** may change the view of building **902** based on the user selection (e.g., to view a selected component, to hide components, etc.).

Referring now to FIG. 10, another integrated BAS-BIM viewer interface **1000** is shown, according to an exemplary embodiment. Interface **1000** may be generated by integrated BAS-BIM viewer **504** to view and interact with a BIM with integrated BAS points and values. Interface **1000** shows a view of building **902** from a location within building **902**. Interface **1000** is shown to include structural components of building **902** such as walls **1006**, floor **1008**, ceiling **1010**, and doors **1012**. Interface **1000** also displays HVAC components **1014** (e.g., air ducts), lighting components **1016**

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(e.g., lighting fixtures), electronic components **1018** (e.g., computer monitors), and furniture **1020** (e.g., desks). The geometry and locations of components **1006-1020** may be defined by the BIM objects within the BIM model.

Interface **1000** is shown to include an object tree **1002**. Object tree **1002** may be displayed in response to selecting tree button **1004**. Object tree **1002** includes a hierarchical representation of building **902** and the various spaces and components contained therein. For example, object tree **1002** is shown to include objects representing a campus, a particular building within the campus (e.g., Jolly Board Tower), levels within the building (e.g., level 0, level 1, level 2, etc.), and spaces/components within each level (e.g., conference rooms, offices, AHUs, etc.). Selecting any of the objects displayed in object tree **1002** may cause interface **1000** to display the selected object or change the view to show the selected object.

Referring now to FIG. 11, another integrated BAS-BIM viewer interface **1100** is shown, according to an exemplary embodiment. Interface **1100** may be generated by integrated BAS-BIM viewer **504** to view and interact with a BIM with integrated BAS points and values. Interface **1100** is shown to include the same view of building **902** as shown in interface **1000**. The view can be changed by selecting view button **1102**. For example, selecting view button **1102** may cause view menu **1116** to be displayed. View menu **1116** is shown to include a perspective button **1104**, a side button **1106**, a top button **1108**, and a front button **1110**. Selecting any of buttons **1104-1110** may cause the view to change to the selected view. Selecting activity button **1118** may allow a user to change the view to simulate the perspective of a person walking through building **902**.

Interface **1110** is shown to include a filter menu **1114**. Filter menu **1114** may be displayed in response to selecting filter button **1112**. Filter menu **1114** includes several categories of objects which can be selectively filtered by checking or unchecking the boxes associated with each category. For example, filter menu **1114** is shown to include the categories of architecture, HVAC, lighting, and plumbing. As shown in FIG. 11, unchecking the box associated with the HVAC category causes the HVAC components **1014** to be hidden.

Referring now to FIG. 12, a BIM viewer interface **1200** is shown, according to an exemplary embodiment. Interface **1200** may be generated by integrated BAS-BIM viewer **504** to view and interact with a BIM. Interface **1200** is shown to include a view of an air handling unit (AHU) **1202** within building **902**. In some embodiments, the view shown in FIG. 12 is displayed in response to selecting an object associated with AHU **1202** via object tree **1002**. For example, the object named "Outdoor AHU—Horizontal 1:6 Square Feet of Coil" can be selected via object tree **1002** to cause AHU **1202** to be displayed.

Interface **1200** shows a view of AHU **1202** before the BAS points are integrated with the BIM. For example, selecting or hovering over AHU **1202** may cause information window **1204** to be displayed. Since no BAS points are yet associated with AHU **1202**, information window **1204** displays only the node name of AHU **1202**. Once BAS points are integrated with the BIM, information window **1204** may be modified to display any BAS points/values that have been mapped to the BIM object representing AHU **1202**.

Referring now to FIG. 13, another integrated BAS-BIM viewer interface **1300** is shown, according to an exemplary embodiment. Interface **1300** may be generated by integrated BAS-BIM viewer **504** to view and interact with a BIM with integrated BAS points and values. Interface **1300** shows a

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view of AHU 1202 after BAS points have been integrated with the BIM. In addition to the information shown in interface 1200, information window 1204 is shown to include several BAS points and the present values associated with each BAS point. The point names may be stored as attributes of the BIM and loaded when the BIM is viewed. The present values may be retrieved from the BAS network and displayed within information window 1204 when AHU 1202 is selected.

Interface 1300 is shown to include a BAS information window 1302. Window 1302 is shown to include several types of information retrieved from the BAS. For example, window 1302 is shown to include BAS points 1304 that have been mapped to AHU 1202, an EFIRM link 1306, technical details about AHU 1202 (e.g., a product data sheet, a catalogue, drawings, etc.), and work order information 1310 describing any work orders that have been performed or scheduled for AHU 1202. Any faults associated with BAS points 1304 or AHU 1202 may be displayed in BAS information window 1302. For example, if the BAS points “HTG-O” and “WC-ADJ” are out of range or otherwise indicate a fault condition, these BAS points may be highlighted in window 1302 (e.g., by coloring portion 1312 red or flashing the BAS points or present values, etc.). Advantageously, interface 1300 allows a user to see not only the BIM information, but also integrated BAS information on a single display.

Referring now to FIG. 14, another integrated BAS-BIM viewer interface 1400 is shown, according to an exemplary embodiment. Interface 1400 may be generated by integrated BAS-BIM viewer 504 to view and interact with a BIM with integrated BAS points and values. Interface 1400 is shown to include a plurality of BAS points 1418 associated with AHU 1202 in object tree 1002. BAS points 1418 may include any point from the BAS that has been mapped to the object associated with AHU 1202 (e.g., a humidity measurement, a temperature measurements, a temperature setpoint, etc.).

Interface 1400 is shown to include a point information window 1402. Point information window 1402 may be displayed in response to selecting a BAS point 1408 in object tree 1002. For example, point information window 1402 may be displayed when the BAS point “FEC.ZN-H” is selected in object tree 1002. Point information window 1402 is shown to include a trend data portion 1404. Trend data portion 1404 may include a graph 1406 of past values of the selected BAS point within a user-defined time range. A user can define the time range for which trend data is displayed by entering times via text boxes 1412. Graph 1406 may include a history of past values and can be selected to display the value of the BAS point at any instant in time.

Point information window 1402 may include an alarm limits portion 1408 and an alarm history portion 1410. Alarm limits portion 1408 may allow a user to define alarm limits for the BAS point. If the BAS point does not fall within the alarm limits, the BAS point may be indicated as a fault. Alarm history portion 1410 may allow a user to view a history of alarms associated with the BAS point.

Point information window 1402 may allow a user to write new values for the BAS point. For example, point information window 1402 is shown to include a text box 1416 which can be used to enter a user-defined value for the BAS point. Selecting write button 1414 may send the user-defined value to the BAS. This feature may be useful for adjusting a setpoint or calibrating a BAS point.

Referring now to FIG. 15, another integrated BAS-BIM viewer interface 1500 is shown, according to an exemplary

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embodiment. Interface 1500 may be generated by integrated BAS-BIM viewer 504 to view and interact with a BIM with integrated BAS points and values. Interface 1500 is shown to include an alarms summary window 1504. Window 1504 may be displayed in response to selecting alarms button 1502. Alarms summary window 1504 is shown to include an indication of any alarms faults associated with the BAS points. Window 1504 may describe each alarm by identifying BAS point associated with the alarm (e.g., by point name), an alarm status (e.g., high or low), a description of the alarm (e.g., heating output), an item reference associated with the alarm, and a device type for the associated BAS point. In some embodiments, alarms summary window 1504 provides user interface options to perform automated fault diagnostics to detect an underlying fault associated with the alarms or to respond to alarms. Selecting an alarm in alarms summary window 1504 may change the view of the building so that the object associated with the alarm is shown (e.g., navigating to a portion of the building that includes the object, zooming in on the object, etc.). The object associated with the alarm may be highlighted in the view of the building (e.g., colored red, flashing, etc.) so that the user can easily identify the object in the view window.

Referring now to FIG. 16, another integrated BAS-BIM viewer interface 1600 is shown, according to an exemplary embodiment. Interface 1600 may be generated by integrated BAS-BIM viewer 504 to view and interact with a BIM with integrated BAS points and values. Interface 1600 is shown to include a trend window 1602, which may be displayed in response to a user selecting trend button 1610. Trend window 1602 may be configured to display trend data for multiple BAS points on the same graph. For example, trend window 1602 is shown displaying trend data for the BAS points “FEC.ZN-Q” and “FEC.WC-ADJ” concurrently on the same graph. Advantageously, trend window 1602 may display trend data (e.g., time series data) for all of the BAS points mapped to a particular BIM object in a single display without requiring any additional user input or configuration to identify the BAS points. BAS points can be displayed or removed from the graph by selecting or deselecting point labels 1608. In some embodiments, interface 1600 includes a time range selector 1606 which allows a user to define the start time and the end time for the trend data displayed in trend window 1602. In some embodiments, interface 1600 includes a value display box 1604 which displays values for one or more of the BAS points shown in the graph at a particular instant in time. A user can select or hover over a portion of the graph to specify the instant in time for which the data values are displayed.

Referring now to FIG. 17, another integrated BAS-BIM viewer interface 1700 is shown, according to an exemplary embodiment. Interface 1700 may be generated by integrated BAS-BIM viewer 504 to view and interact with a BIM with integrated BAS points and values. Interface 1700 is shown to include a search box 914. Search box 914 can be used to search for particular BIM objects, BAS points, particular rooms or zones, building equipment, or other objects or data points that match a user-defined search term. Interface 1700 may be configured to search BAS point names, BIM object names, BIM object attributes, or other items included in the integrated BIM model. Results of the search may be displayed in search results list 1702. Selecting an item in search results list 1702 may change the view of building 902 based on the user selection (e.g., to view a selected object or an object associated with a selected BAS point).

Referring now to FIG. 18, a point mapping interface 1800 is shown, according to an exemplary embodiment. Interface

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1800 may be generated by integrated BAS-BIM integrator 502 to map BAS points to BIM objects. Interface 1800 may allow a user to identify a BIM or upload a BIM (e.g., by selecting upload button 1806). The uploaded BIM may be used to generate a BIM tree 1804, which may include a hierarchical listing of BIM objects. Interface 1800 may automatically identify a corresponding BAS and retrieve a BAS tree 1802 from the BAS network. Interface 1800 may allow a user to rename the identified BAS and/or identify a different BAS (e.g., by entering a campus name 1808, building name 1810, etc.). The identified BAS may be used to generate a BAS tree 1802, which may include a hierarchical listing of BAS points. A user can define a mapping between BAS points and BIM objects by dragging and dropping BAS points from BAS tree 1802 onto BIM objects in BIM tree 1804. In some embodiments, the point mappings are stored in a point mappings database. In other embodiments, the mapped BAS points are stored as attributes or properties of the BIM object to which the BAS points are mapped.

Changes to the building or point mappings can be made by uploading a new BIM. For example, if a BAS device is moved from one room in the building to another room in the building, an updated BIM reflecting the change can be uploaded via point mapping interface 1800. Point mapping interface 1800 may be configured to retrieve a previous point mapping from the point mappings database and automatically apply the point mappings to the updated BIM (e.g., by selecting "keep record" button 1812). Advantageously, this feature allows the point mappings to be updated and applied to new versions of the BIM without requiring a user to redefine the point mappings.

BAS-BIM Integration Process

Referring now to FIG. 19, a flowchart of a process 1900 for generating and using a BIM with integrated BAS points is shown, according to an exemplary embodiment. Process 1900 may be performed by one or more components of system 500, system 700, or controller 800, as described with reference to FIGS. 5-8.

Process 1900 is shown to include receiving a building information model (BIM) (step 1902). The BIM may include a plurality of BIM objects representing building equipment. In some embodiments, the BIM includes a three-dimensional model of the building. The BIM objects may include one or more objects representing structural components of the building and one or more objects representing spaces within the building.

Process 1900 is shown to include collecting building automation system (BAS) points from a BAS network (step 1904). The BAS network may include a BACnet network, a LonWorks network, or any other network configured to facilitate communications between building equipment. The BAS points may be measured data points, calculated data points, setpoints, or other types of data points used by the BAS, generated by the BAS, or stored within the BAS (e.g., configuration settings, control parameters, equipment information, alarm information, etc.). In some embodiments, step 1904 includes retrieving corresponding point values from the BAS network. The point values may include at least one of values measured by the building equipment, values generated by the building equipment, setpoints for the building equipment, and operating parameters for the building equipment.

Process 1900 is shown to include integrating the BAS points with the BIM (step 1906). In some embodiments, step 1906 includes generating a BAS tree that includes the BAS points, generating a BIM tree that includes the BIM objects,

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and generating a mapping interface that includes the BAS tree and the BIM tree. Step 1906 may include establishing mappings between the BAS points and the BIM objects based on a user input received via the mapping interface. For example, the user input may include dragging and dropping the BAS points from the BAS tree onto BIM objects in the BIM tree. In some embodiments, step 1906 includes storing mappings between the BAS points and the BIM objects in a mappings database.

Process 1900 is shown to include using the BIM with the integrated BAS points to generate a user interface including a graphical representation of the BIM objects and the BAS points (step 1908). Several examples of user interfaces that may be generated in step 1908 are described with reference to FIGS. 9-18. The user interface may be viewed using an integrated BAS-BIM viewer (e.g., CAD software, a CAD viewer, a web browser, etc.). The BAS-BIM viewer uses the geometric and location information from the BIM to generate 3D representations of physical components and building spaces. Advantageously, a user can view real-time data from the BAS and/or trend data for objects represented in the BIM simply by viewing the BIM with integrated BAS data.

Process 1900 is shown to include detecting a control action received via the user interface (step 1910) and using the control action to generate a control signal for the building equipment (step 1912). In some embodiments, the user interface is an interactive interface that allows the user to view BAS points, change the values of BAS points (e.g., setpoints), configure the BAS, and/or interact with the BAS via the user interface. For example, the user can write new values for any of the BAS points displayed in the BIM (e.g., setpoints), send operating commands or control signals to the building equipment displayed in the BIM, or otherwise interact with the BAS via the BIM.

Control actions submitted via the user interface may be provided to the BAS network. The BAS network may use the control actions to generate control signals for the building equipment or otherwise adjust the operation of the building equipment. In this way, the BIM with integrated BAS points and values not only allows a user to monitor the BAS, but also provides the control functionality of a graphical BAS management and control interface. These features allow the BIM with integrated BAS data to be used as a building control interface which provides a graphical 3D representation of the building and the equipment contained therein without requiring a user to manually create or define graphics for various building components.

Configuration of Exemplary Embodiments

The construction and arrangement of the systems and methods as shown in the various exemplary embodiments are illustrative only. Although only a few embodiments have been described in detail in this disclosure, many modifications are possible (e.g., variations in sizes, dimensions, structures, shapes and proportions of the various elements, values of parameters, mounting arrangements, use of materials, colors, orientations, etc.). For example, the position of elements may be reversed or otherwise varied and the nature or number of discrete elements or positions may be altered or varied. Accordingly, all such modifications are intended to be included within the scope of the present disclosure. The order or sequence of any process or method steps may be varied or re-sequenced according to alternative embodiments. Other substitutions, modifications, changes, and omissions may be made in the design, operating conditions

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and arrangement of the exemplary embodiments without departing from the scope of the present disclosure.

The present disclosure contemplates methods, systems and program products on any machine-readable media for accomplishing various operations. The embodiments of the present disclosure may be implemented using existing computer processors, or by a special purpose computer processor for an appropriate system, incorporated for this or another purpose, or by a hardwired system. Embodiments within the scope of the present disclosure include program products comprising machine-readable media for carrying or having machine-executable instructions or data structures stored thereon. Such machine-readable media can be any available media that can be accessed by a general purpose or special purpose computer or other machine with a processor. By way of example, such machine-readable media can comprise RAM, ROM, EPROM, EEPROM, CD-ROM or other optical disk storage, magnetic disk storage or other magnetic storage devices, or any other medium which can be used to carry or store desired program code in the form of machine-executable instructions or data structures and which can be accessed by a general purpose or special purpose computer or other machine with a processor. Combinations of the above are also included within the scope of machine-readable media. Machine-executable instructions include, for example, instructions and data which cause a general purpose computer, special purpose computer, or special purpose processing machines to perform a certain function or group of functions.

Although the figures show a specific order of method steps, the order of the steps may differ from what is depicted. Also two or more steps may be performed concurrently or with partial concurrence. Such variation will depend on the software and hardware systems chosen and on designer choice. All such variations are within the scope of the disclosure. Likewise, software implementations could be accomplished with standard programming techniques with rule based logic and other logic to accomplish the various connection steps, processing steps, comparison steps and decision steps.

The background section is intended to provide a background or context to the invention recited in the claims. The description in the background section may include concepts that could be pursued, but are not necessarily ones that have been previously conceived or pursued. Therefore, unless otherwise indicated herein, what is described in the background section is not prior art to the present invention and is not admitted to be prior art by inclusion in the background section.

What is claimed is:

1. A building system of a building comprising one or more storage media having instructions stored thereon that, when executed by one or more processors, cause the one or more processors to:

generate a plurality of mappings between a plurality of points of a building automation system (BAS) network for a plurality of pieces of building equipment and a plurality of graphic representations of the plurality of pieces of building equipment of a building information model (BIM), wherein the plurality of mappings map a point of the plurality of points of a piece of building equipment of the plurality of pieces of building equipment with a graphic representation of the piece of building equipment of the plurality of graphic representations;

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cause a user device to display an interface comprising:

a graphic element comprising point names of the plurality of points mapped to the plurality of pieces of building equipment, wherein the graphic element comprises a hierarchical representation of the building, the hierarchical representation comprising a plurality of levels, wherein:

a first level of the plurality of levels includes an indication of a floor of the building;

a second level that depends from the first level includes an indication of the piece of building equipment, wherein the second level depending from the first level indicates that the piece of building equipment is located on the floor; and

a third level depending from the second level including an indication of the point, wherein the third level depending from the second level indicates that the piece of building equipment includes the point; and

a view of the BIM including at least one of the plurality of graphic representations of the plurality of pieces of building equipment;

receive, via the graphic element, a selection of the point of the plurality of points, the point mapped to the graphic representation of the piece of building equipment; and

modify, based on the plurality of mappings and the selection, the view of the BIM to display the graphic representation of the piece of building equipment mapped to the point.

2. The building system of claim 1, wherein the graphic element comprises a plurality of rows representing the plurality of points, wherein a row of the plurality of rows comprises:

a name of the point; and

an alarm indicator for the point, the alarm indicator based on values of the point;

wherein the instructions cause the one or more processors to:

receive, via the graphic element, the selection of the point of the plurality of points by receiving the selection of the row.

3. The building system of claim 1, wherein the instructions cause the one or more processors to:

receive, via the graphic element, the selection of the point of the plurality of points by receiving the selection of the third level.

4. The building system of claim 1, wherein the graphic element is located below the view of the BIM within the interface.

5. The building system of claim 1, wherein the graphic element is located on a left side or a right side of the view of the BIM within the interface.

6. The building system of claim 1, wherein the instructions cause the one or more processors to:

store technical detail information for the piece of building equipment in the one or more storage media; and

cause the view of the BIM to include the technical detail information in a particular location within the view of the BIM associated with the graphic representation of the piece of building equipment responsive to the selection.

7. The building system of claim 1, wherein the instructions cause the one or more processors to:

generate, based on at least a portion of the plurality of mappings, an alarm for the piece of building equipment based on a plurality of values of the point of the

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plurality of points mapped to the piece of building equipment by the portion of the plurality of mappings and an alarm rule for the piece of building equipment.

8. The building system of claim 1, wherein the instructions cause the one or more processors to:

receive, via the user device, a configuration of an alarm rule, the alarm rule and the configuration defining generation of an alarm for the piece of building equipment; and

generate, based on at least a portion of the plurality of mappings, the alarm for the piece of building equipment based on the configuration of the alarm rule.

9. The building system of claim 1, wherein the instructions cause the one or more processors to:

retrieve, based on the plurality of mappings, a plurality of values of the point from the one or more storage media responsive to receiving the selection; and

generate data causing the interface to include an element comprising a trend for the point based on the plurality of values of the point.

10. The building system of claim 1, wherein the instructions cause the one or more processors to:

generate a window responsive to the selection, the window comprising:

a graph including a trend line of a plurality of values of the point, the graph comprising a vertical axis indicating the plurality of values and a horizontal axis indicating time; and

a time range element providing a selection of a time range from a starting time to an ending time;

receive, via the time range element of the window, a particular time range including a particular starting time and a particular ending time; and

cause the graph to display the trend line from the particular starting time to the particular ending time.

11. A method comprising:

generating, by one or more processing circuits, a plurality of mappings between a plurality of points of a building automation system (BAS) network of a building for a plurality of pieces of building equipment and a plurality of graphic representations of the plurality of pieces of building equipment of a building information model (BIM), wherein the plurality of mappings map a point of the plurality of points of a piece of building equipment of the plurality of pieces of building equipment with a graphic representation of the piece of building equipment of the plurality of graphic representations;

causing, by the one or more processing circuits, a user device to display an interface comprising:

a graphic element comprising point names of the plurality of points mapped to the plurality of pieces of building equipment, wherein the graphic element comprises a hierarchical representation of the building, the hierarchical representation comprising a plurality of levels, wherein:

a first level of the plurality of levels includes an indication of a floor of the building;

a second level that depends from the first level includes an indication of the piece of building equipment, wherein the second level depending from the first level indicates that the piece of building equipment is located on the floor; and

a third level depending from the second level including an indication of the point, wherein the third level depending from the second level indicates that the piece of building equipment includes the point; and

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a view of the BIM including at least one of the plurality of graphic representations of the plurality of pieces of building equipment;

receiving, by the one or more processing circuits, via the graphic element, a selection of the point of the plurality of points, the point mapped to the graphic representation of the piece of building equipment; and

modifying, by the one or more processing circuits, based on the plurality of mappings and the selection, the view of the BIM to display the graphic representation of the piece of building equipment mapped to the point.

12. The method of claim 11, wherein the graphic element comprises a plurality of rows representing the plurality of points, wherein a row of the plurality of rows comprises:

a name of the point; and

an alarm indicator for the point, the alarm indicator based on values of the point;

wherein the method comprises:

receiving, by the one or more processing circuits, via the graphic element, the selection of the point of the plurality of points by receiving the selection of the row.

13. The method of claim 11,

wherein receiving, by the one or more processing circuits, via the graphic element, the selection of the point of the plurality of points comprises receiving the selection of the third level.

14. The method of claim 11, further comprising:

storing, by the one or more processing circuits, technical detail information for the piece of building equipment in one or more storage media; and

causing, by the one or more processing circuits, the view of the BIM to include the technical detail information in a particular location within the view of the BIM associated with the graphic representation of the piece of building equipment responsive to the selection.

15. The method of claim 11, further comprising:

generating, by the one or more processing circuits, based on at least a portion of the plurality of mappings, an alarm for the piece of building equipment based on a plurality of values of the point of the plurality of points mapped to the piece of building equipment by the portion of the plurality of mappings and an alarm rule for the piece of building equipment.

16. The method of claim 11, further comprising:

receiving, by the one or more processing circuits, via the user device, a configuration of an alarm rule, the alarm rule and the configuration defining generation of an alarm for the piece of building equipment; and

generating, by the one or more processing circuits, based on at least a portion of the plurality of mappings, the alarm for the piece of building equipment based on the configuration of the alarm rule.

17. The method of claim 11, further comprising:

retrieving, by the one or more processing circuits, based on the plurality of mappings, a plurality of values of the point from one or more storage media responsive to receiving the selection; and

generating, by the one or more processing circuits, data causing the interface to include an element comprising a trend for the point based on the plurality of values of the point.

18. The method of claim 11, further comprising:

generating, by the one or more processing circuits, a window responsive to the selection, the window comprising:

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a graph including a trend line of a plurality of values of the point, the graph comprising a vertical axis indicating the plurality of values and a horizontal axis indicating time; and

a time range element providing a selection of a time range from a starting time to an ending time;

receiving, by the one or more processing circuits, via the time range element of the window, a particular time range including a particular starting time and a particular ending time; and

causing, by the one or more processing circuits, the graph to display the trend line from the particular starting time to the particular ending time.

19. One or more storage media having instructions stored thereon that, when executed by one or more processors, cause the one or more processors to:

generate a plurality of mappings between a plurality of points of a building automation system (BAS) network of a building for a plurality of pieces of building equipment and a plurality of graphic representations of the plurality of pieces of building equipment of a building information model (BIM), wherein the plurality of mappings map a point of the plurality of points of a piece of building equipment of the plurality of pieces of building equipment with a graphic representation of the piece of building equipment of the plurality of graphic representations;

cause a user device to display an interface comprising:

a graphic element comprising point names of the plurality of points mapped to the plurality of pieces of building equipment, wherein the graphic element comprises a hierarchical representation of the building, the hierarchical representation comprising a plurality of levels, wherein:

a first level of the plurality of levels includes an indication of a floor of the building;

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a second level that depends from the first level includes an indication of the piece of building equipment, wherein the second level depending from the first level indicates that the piece of building equipment is located on the floor; and

a third level depending from the second level including an indication of the point, wherein the third level depending from the second level indicates that the piece of building equipment includes the point; and

a view of the BIM including at least one of the plurality of graphic representations of the plurality of pieces of building equipment;

receiving, via the graphic element, a selection of the point of the plurality of points, the point mapped to the graphic representation of the piece of building equipment; and

modify, based on the plurality of mappings and the selection, the view of the BIM to display the graphic representation of the piece of building equipment mapped to the point.

20. The one or more storage media of claim **19**, wherein the graphic element comprises a plurality of rows representing the plurality of points, wherein a row of the plurality of rows comprises:

a name of the point; and

an alarm indicator for the point, the alarm indicator based on values of the point;

wherein the instructions cause the one or more processors to:

receive, via the graphic element, the selection of the point of the plurality of points by receiving the selection of the row.

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